
GREENCERT: Signed Green Operators for Certified Neural Training Transitions

Ian Rhee

Abstract

Local models can predict neural-training transitions accurately while saying nothing about whether their extrapolation remains valid. A forecast does not become a proof merely because it happens to be right. We present GREENCERT, an anchor-preserving verifier that either brackets a future persistent event or abstains. Starting from a checkpoint-local reference path, it propagates the realized signed defect through a finite-window Green operator, recenters on that directional response, and bounds only the remaining second-order error. The response-centered theorem therefore certifies the realized nonlinear trajectory over a finite window and transports its state tube to a persistent first-passage bracket. Across four frozen or outcome-sealed studies, all 83 issued brackets contain their subsequently revealed crossings. Independent 192-bit outward continuation re-certifies all 63 WDBC and digits brackets. Retaining direction matters: signed propagation preserves a 147-step digits certificate and eight of nine fixed-radius Transformer certificates that the matched unsigned construction loses, while both inaccurate finite digits forecasts abstain before randomized probing. A frozen directional-block remainder audit further shrinks the scalar fourth-order term by $2,899\times$ at the holdout median and recovers three sealed holdout certificates one variational sweep earlier. Matrix-free computational refinements substantially reduce Green and output-operator work without changing the issued brackets. The resulting verifier is selective and explicit about its numerical and probabilistic obligations.

1 Introduction

Training curves contain events that matter in practice: a held-out gate is crossed, a capability appears, or a long plateau ends. Local approximations often predict these events well, but they do not provide a stopping rule for their own extrapolation. Conventional norm bounds have the opposite problem. They pay for every admissible direction, including directions the realized defect never takes, and can become vacuous while the local path remains accurate.

We study a sharper object: a *checkpoint-conditioned first-passage certificate*. For a fixed optimizer trajectory and an evaluation gate, the method returns an interval containing the first future persistent crossing or explicitly abstains. This is a deterministic event statement on a fixed evaluation set, not a population-generalization bound. A local clock proposes an event time. GREENCERT additionally checks the realized finite-window dynamics and either returns a persistent first-passage bracket or declines to certify the proposal. The main result is a finite-window shadowing and event-transport theorem. The experiments test three separate questions: when the theorem issues, whether an issued bracket covers, and what the certificate costs to construct.

Grokking makes the distinction vivid. Mechanistic progress measures can reveal latent structure [Nanda et al., 2023], and recent work predicts delayed generalization from loss geometry or calibrated clocks [Notsawo et al., 2023, Kim, 2026, Khanh et al., 2026]. None of these predictions by itself certifies the future nonlinear trajectory. Conversely, shadowing theory can validate pseudo-orbits, but classical formulations do not produce neural output-event brackets from a fixed training checkpoint.

Our construction begins with an anchor-fixed local quadratic optimizer clock and improves it by causal variational sweeps. At the final reference path $c_0, \dots, c_H$, we form its signed defect $s_j = G(c_j) - c_{j+1}$ and ordered response $z = K_H s$. We then recenter the unknown trajectory error on z : the known first-order displacement is preserved with its sign and time ordering, while only a second-order remainder is exposed to worst-case gain. This response-centered closure avoids charging the full first-order quantity $\|K_H\| \|s\|$, or the complete displacement $\|z\|$, to an unknown remainder.

Our contributions are:

1. We formulate training-event certification as an anchor-fixed causal sequence problem and derive a response-centered Green theorem with explicit implementation error. A time-resolved corollary aligns checkpointwise drift with the signed response, while corrected-path relinearization removes the mixed state-response term entirely under the same bounded-Hessian envelope. A polarized block-majorant corollary retains the three realized directions in the fourth-order gradient remainder and maximizes only over the remaining dual direction. For scaled momentum, a forcing-subspace corollary closes the nonlinear problem in parameter space and removes optimizer-state directions that neither generate the nonlinear remainder nor affect the neural event.
2. We give a matrix-free, anytime implementation. HVP/VJP products apply K_H and $K_H^\top$; one Gaussian block bounds every inspected Gram power, while predictable spending permits adaptive operator creation. Verified residual identities and operator-cap budgets control inexact products and recentering; fresh probes can bypass a second response or certify its Green image using outward scalar directional jets.
3. We run frozen confirmations on three task families and two separately sealed Transformer cohorts. All 83 issued brackets cover; every one of the 63 non-modular brackets survives independent 192-bit continuation, and both inaccurate finite digits clocks are rejected before randomized probing.
4. We isolate the roles of direction and computation. A matched unsigned construction retains only one of nine Transformer certificates and loses the sealed 147-step digits event. We develop corrected-path relinearization, deterministic neural-jet release, prefix streaming, and batched operator products to reduce certificate cost without changing issued brackets. In a frozen 15-case audit, these refinements retain all 15 brackets while reducing Green applications from 560 to 64 ($8.75\times$). In a separate frozen sweep-minimization audit, the directional remainder recovers three holdout certificates under a three-sweep reference that the scalar remainder cannot close.

2 Problem and local clock

Let an optimizer, including any auxiliary state, define a differentiable map

$$x_{t+1} = G(x_t), \quad x_t \in \mathbb{R}^d. \quad (1)$$

For ordinary gradient descent $x = \theta$; for momentum we augment parameters with velocity. At anchor x_a , the raw local clock is the affine iteration

$$c_0^{(0)} = x_a, \quad c_{j+1}^{(0)} = G(x_a) + DG(x_a)(c_j^{(0)} - x_a). \quad (2)$$

For gradient descent this is exactly the complete local quadratic model $d_{j+1} = (I - \eta H_a)d_j - \eta g_a$. It can be propagated with HVPs and does not require selecting or truncating Hessian eigenmodes.

We improve a reference path by a causal variational sweep. For sweep k ,

$$r_j^{(k)} = G(c_j^{(k)}) - c_{j+1}^{(k)}, \quad z_0^{(k)} = 0, \quad z_{j+1}^{(k)} = DG(c_j^{(k)})z_j^{(k)} + r_j^{(k)}, \quad c_j^{(k+1)} = c_j^{(k)} + z_j^{(k)}. \quad (3)$$

This is an anchor-preserving Newton/pseudo-orbit correction. Let $\kappa_{\infty, H}$ be the induced gain from the maximum injection norm to the maximum state norm for this causal variational recurrence. If DG is M -Lipschitz throughout every segment joining a path point to its corrected point, one sweep contracts the maximum path defect quadratically:

$$\max_j \|r_j^{(k+1)}\|_2 \leq \frac{1}{2} M \kappa_{\infty, H}^2 \left(\max_i \|r_i^{(k)}\|_2 \right)^2. \quad (4)$$

This quadratic mechanism survives approximate arithmetic. Appendix Proposition 4 gives the exact inexact-sweep identity $r'_j = N_j(\tilde{z}_j) + (r_j - \tilde{r}_j) - d_j$, so verified defect and recurrence errors enter additively rather than replacing the directional correction by a norm bound. Each protocol freezes its sweep count (three for digits, four otherwise). The signed response below is *not* an adaptive extra sweep; it encloses error around the already frozen final centerline.

For evaluation examples $i = 1, \dots, N$, let n_j be the number correctly classified at future update $a + j$. A gate requires $n_j \geq r$. Its unrestricted K -persistent first passage is

$$T_{r,K} = \inf\{j \in \mathbb{N}_0 : n_{j+\ell} \geq r \text{ for every } 0 \leq \ell < K\}, \quad (5)$$

with $\inf \emptyset = \infty$. On a certified horizon H , endpoint searches are restricted to $0 \leq j \leq H - K + 1$, so every reported endpoint uses a complete persistence block. If either endpoint cannot be established in this range, the method abstains. The scientific object is a bracket $[L, U]$ guaranteed to contain $T_{r,K}$, or abstention.

3 Signed finite-window certification

3.1 The Green closure

Fix any computable reference $c_0, \dots, c_H$ with $c_0 = x_a$ and define

$$s_j = G(c_j) - c_{j+1}, \quad J_j = DG(c_j). \quad (6)$$

For injections $u = (u_0, \dots, u_{H-1})$, the causal Green operator K_H is defined by

$$z_0 = 0, \quad z_{j+1} = J_j z_j + u_j, \quad K_H u = (z_1, \dots, z_H). \quad (7)$$

We equip sequence space with $\|z\|_X^2 = \sum_{j=1}^H \|z_j\|_2^2$ and injection space with $\|u\|_U^2 = \sum_{j=0}^{H-1} \|u_j\|_2^2$. Unlike products of scalar one-step bounds, $K_H : U \rightarrow X$ retains every ordered, signed, time-varying propagator.

Let

$$N_j(u) = G(c_j + u) - G(c_j) - J_j u. \quad (8)$$

For the realized anchor-fixed error $h_j = x_{a+j} - c_j$, causal substitution gives the exact identity

$$h = K_H s + K_H N(h), \quad h_0 = 0. \quad (9)$$

This identity separates the part of the trajectory error that is known from the part that remains to be certified. The term $K_H s$ is the signed first-order displacement caused by the realized reference-path defects, while $K_H N(h)$ is the unknown nonlinear remainder. Rather than bounding the whole error in a tube centered at zero, GREENCERT centers the tube at a computed approximation to $K_H s$. The theorem asks whether the remaining nonlinear response fits inside a radius E while the full tube stays within the domain where the derivative bound is valid.

Theorem 1 (Response-centered finite-window certificate). *Assume that, for $1 \leq j < H$ and $\|u\|_2 \leq \rho$,*

$$\|DG(c_j + u) - J_j\|_2 \leq M \|u\|_2. \quad (10)$$

Let $\kappa \geq \|K_H\|$ and let a computed signed response $\tilde{z}$, with $\tilde{z}_0 = 0$, satisfy

$$\|K_H s - \tilde{z}\|_X \leq \alpha. \quad (11)$$

Set $Z = \|\tilde{z}\|_X$, $p = \max_j \|\tilde{z}_j\|_2$, and $q_j = N_j(\tilde{z}_j)$ for $0 \leq j < H$ (with $q_0 = 0$). Suppose $\beta \geq \|K_H q\|_X$, and write $b = \kappa M$ and $Y = \alpha + \beta$. If some $E \geq 0$ obeys

$$p + E \leq \rho, \quad Y + bpE + \frac{b}{2}E^2 \leq E, \quad (12)$$

then the realized trajectory obeys

$$\|h - \tilde{z}\|_X \leq E, \quad \|h_j\|_2 \leq p + E \quad (1 \leq j \leq H). \quad (13)$$

The exact anchor makes the omitted $j = 0$ drift condition rigorous: $h_0 = \tilde{z}_0 = 0$ implies $N_0(h_0) = N_0(\tilde{z}_0) = 0$. State c_H is not a transition input. More explicitly, suppose $\|s - \tilde{s}\|_U \leq \sigma$ and the computed response has recurrence residual $d_j = \tilde{z}_{j+1} - J_j \tilde{z}_j - \tilde{s}_j$ with $\|d\|_U \leq \tau$. Since

$\tilde{z} = K_H(\tilde{s} + d)$ exactly, one may take $\alpha = \kappa(\sigma + \tau)$. Thus local verified residuals supply the outward response interface in Eq. (11). If $\tilde{q}$ approximates $q = N(\tilde{z})$ and an anchor-fixed $\tilde{y}$ has recurrence residual bounded by τ_q , the same identity gives

$$\beta = \|\tilde{y}\|_X + \kappa(\sigma_q + \tau_q), \quad \|q - \tilde{q}\|_U \leq \sigma_q. \quad (14)$$

Appendix Corollary 6 proves this two-response interface, so neither causal response requires exact arithmetic. Directly subtracting optimizer maps can be ill-conditioned when q is quadratically small. Appendix Corollary 7 instead forms $D^2G(c_j)[\tilde{z}_j, \tilde{z}_j]/2$ directionally and encloses its explicit cubic remainder. That scalar enclosure still forgets how the known parameter correction is distributed across the network. Appendix Corollary 8 carries a nonnegative fourth-order block majorant and leaves only one dual direction worst-case.

Corollary 1 (Zero-extra-operator response radius). *Taylor’s theorem gives $\|q\|_U \leq MpZ/2$, so Theorem 1 may use $\beta = \kappa MpZ/2$ without an additional model evaluation or operator query. With $Y = \alpha + bpZ/2$, closure exists when*

$$bp < 1, \quad D = (1 - bp)^2 - 2bY \geq 0, \quad E = \frac{2Y}{1 - bp + \sqrt{D}}, \quad (15)$$

provided $p + E \leq \rho$. The expression for E avoids cancellation when Y is small.

Corrected-path relinearization. For $b = c + v$, rebuilding the Jacobians at b gives the exact defect $\bar{s}_j = N_j(v_j) - r_j^v$ and closure $Y + \bar{\kappa}ME^2/2 \leq E$, eliminating κMpE . Here $E = 2Y/(1 + \sqrt{1 - 2\bar{\kappa}MY})$ under the local domain condition and a pairwise M -Lipschitz Jacobian envelope, supplied by our bounded-Hessian bounds. Appendix Corollary 11 states and proves the result.

Scaled-momentum forcing subspace. For $G(\theta, w) = (\theta - r, r)$ with $r = \mu w + \eta \nabla F(\theta)$, every nonlinear remainder is exactly $\mathcal{B}q = (-\eta q, \eta q)$ and depends only on the parameter error. Writing Π_θ for parameter projection, the corrected closure may therefore query $\Pi_\theta \bar{K}_H \mathcal{B}$ instead of the full $\bar{K}_H$. Appendix Corollary 12 gives the closed parameter-sequence fixed point, including the one-step causal shift at the fixed anchor. With exact quantities, its quadratic coefficient and response term are never larger than the full-state counterparts. In a separately sealed post-release audit over all 15 Green-evaluable operators, the conservative implementation that retains the identically zero first forcing block preserves 15/15 brackets and reduces staged logical Green sweeps $112 \rightarrow 96$ (1.167 $\times$).

Prefix-local staged release. For fixed T and Gaussian probes g_i , the same projection event gives both $\|T\| \leq \max_i \|Tg_i\|/c_\delta$ and $\|T\| \leq (\max_i \|T^T Tg_i\|/c_\delta)^{1/2}$. We first try the direct image and apply T^T only if closure fails; prespecified nested prefixes permit valid stopping without another probability charge. Causal recentering commutes with prefix restriction, so a sealed horizon need not construct later states (Appendix Corollary 14 and Proposition 5).

Deterministic neural-jet release. The ball-valid output jets already used for nonlinear drift can sometimes discharge the event transport as well. If $\|Df_j\| \leq A_j$, $\|D^2f_j\| \leq B_j$, and $\|D^3f_j\| \leq C_j$ on the outer ball, mean cross entropy gives the scaled-momentum drift bound

$$M_{\text{jet}} = \max_{1 \leq j < H} \sqrt{2}\eta \left(2A_j^3 + \frac{3}{2}A_jB_j + \sqrt{2}C_j \right). \quad (16)$$

After corrected-path closure, a true-class-versus-competitor margin moves by at most $\sqrt{2}A_j(p + E)$. Thus this deterministic screen may issue before any output-Jacobian probe; unresolved cases fall back to the sharper randomized operator under the original predictable budget. Appendix Corollary 13 states the release formally.

The earlier fixed-radius rule is recovered by centering the ball at zero: $Z + \kappa MR^2/2 \leq R$; setting $R = 2Z$ gives $2\kappa MZ \leq 1$. We retain this rule as a matched baseline. The response-centered theorem instead pays the signed first-order displacement in the center and exposes only a quadratic residual to the Green gain.

Corollary 2 (Time-resolved drift alignment). *If Eq. (10) holds with checkpointwise constants M_j , let $d_j = \|\tilde{z}_j\|_2$ for $1 \leq j < H$ and define*

$$Q = \frac{1}{2} \left(\sum_{j=1}^{H-1} M_j^2 d_j^4 \right)^{1/2}, \quad A = \max_{1 \leq j < H} M_j d_j, \quad B = \max_{1 \leq j < H} M_j. \quad (17)$$

Then the sufficient closure is

$$\alpha + \kappa Q + \kappa AE + \frac{1}{2}\kappa BE^2 \leq E, \quad p + E \leq \rho. \quad (18)$$

Relative to Corollary 1 with $M := \max_{1 \leq j < H} M_j$, it is never weaker, since $Q \leq MpZ/2$, $A \leq Mp$, and $B = M$.

3.2 A matrix-free, family-wise Green bound

For any matrix-free linear operator T , its Gram operator $A = T^\top T$ is positive semidefinite. For $T = K_H$, a forward product is one sweep of optimizer JVPs and a transpose product is the reverse VJP sweep. Neither forms a Jacobian, Hessian, or propagator matrix; the same construction applies to neural-output Jacobians.

Proposition 1 (Anytime Gaussian Gram enclosure). *For a fixed linear operator T , $0 < \delta < 1$, and one committed block of m independent standard Gaussian vectors g_i , define, for every integer $q \geq 1$,*

$$Y_q = \max_i \|A^q g_i\|, \quad c_\delta = \Phi^{-1}\left(\frac{1 + \delta^{1/m}}{2}\right). \quad (19)$$

Then, in exact arithmetic, with probability at least $1 - \delta$, simultaneously for every inspected power q ,

$$\|T\| \leq (Y_q / c_\delta)^{1/(2q)}. \quad (20)$$

One top-eigenvector projection event implies Eq. (20) for every q , so inspecting powers adaptively incurs no union penalty.

Proposition 2 (Predictable operator-family spending). *Let $\mathcal{F}_{n-1}$ contain all information before operator query n . Choose T_n and $0 < \delta_n < 1$ measurably from that history, then draw a fresh Gaussian block independent of T_n conditional on $\mathcal{F}_{n-1}$. If $\sum_{n \geq 1} \delta_n \leq \delta$ almost surely, every adaptively queried operator and every inspected power is valid simultaneously with probability at least $1 - \delta$.*

The frozen studies use the simpler finite-family version: operator identities are outcome-blind and independent of domain-separated probe streams beneath a sealed OS-random nonce.

Residual-certified products. Exact Gram products are also unnecessary. If computed powers obey $\max_i \|\tilde{v}_{i,\ell+1} - A\tilde{v}_{i,\ell}\| \leq \xi_\ell$ and $\tilde{Y}_q = \max_i \|\tilde{v}_{i,q}\|$, let R_q solve

$$c_\delta R_q^q = \tilde{Y}_q + \sum_{\ell=0}^{q-1} \xi_\ell R_q^{q-1-\ell}. \quad (21)$$

Appendix Proposition 3 gives $\|T\| \leq \sqrt{R_q}$ on the same Gaussian event. Arithmetic precision may be refined after earlier iterates are inspected: valid residual bounds change only this scalar root and consume no additional failure probability (Corollary 4). For a desired squared-norm cap $\bar{R}$, one may instead preallocate $\bar{Y} + \sum_\ell b_\ell \leq c_\delta \bar{R}^q$ and enforce the local checks $\tilde{Y}_q \leq \bar{Y}$ and $\xi_\ell \bar{R}^{q-1-\ell} \leq b_\ell$. Passing certifies $\|T\| \leq \sqrt{\bar{R}}$ directly; a failed product alone can be recomputed at higher precision (Corollary 5).

Output and Green operators may stop at different powers. A stateful verifier therefore evaluates output operators at $q = 1$, advances the Green power until closure, and refines only unresolved quantities. The confidence event is unchanged; unqueried powers consume no HVP/VJP applications.

3.3 Neural derivative and event transport

For scaled momentum state $(\theta, w = \eta v)$,

$$r = \mu w + \eta \nabla F(\theta), \quad G(\theta, w) = (\theta - r, r). \quad (22)$$

The exact products are

$$dr = \mu dw + \eta H d\theta, \quad DG(d\theta, dw) = (d\theta - dr, dr), \quad (23)$$

$$DG^\top(a, b) = (a + \eta H(b - a), \mu(b - a)). \quad (24)$$

If the objective Hessian is L -Lipschitz in parameters, then Eq. (10) holds in scaled state with $M = \sqrt{2}\eta L$. Our block-aware jet propagates first, second, and third parameter derivatives through self-attention, GELU, residual paths, and cross-entropy, with a fixed-point inflation that makes intermediate activation bounds valid on the complete radius- ϱ ball.

Let $m_{iq}(\theta)$ be the true-class logit minus competitor q . Let B_{1j} bound the logit Jacobian at the reference center c_j , and let B_{2j} bound the logit Hessian throughout the certified pointwise parameter ball of radius ϱ about c_j . Then

$$|m_{iq}(x_{a+j}) - m_{iq}(c_j)| \leq \Delta_j := \sqrt{2} (B_{1j}\varrho + \frac{1}{2}B_{2j}\varrho^2). \quad (25)$$

Corollary 3 (Response-centered observable transport). *Write $\bar{c}_j = c_j + \tilde{z}_j$ and $d_j = \|\Pi_\theta \tilde{z}_j\|$. Under the same derivative bounds,*

$$|m_{iq}(x_{a+j}) - m_{iq}(\bar{c}_j)| \leq \Delta_j^{\text{rc}} := \sqrt{2} [(B_{1j} + B_{2j}d_j)E + \frac{1}{2}B_{2j}E^2]. \quad (26)$$

It needs one forward evaluation and no operator query; $d_j \leq p$ gives $\Delta_j^{\text{rc}} \leq \Delta_j$ at $\varrho = p + E$, removing the known response from output uncertainty.

Strict positive lower margins give a guaranteed-correct count n_j^- ; one strict negative upper margin gives definite incorrectness and hence a possible-correct count n_j^+ . Define

$$\begin{aligned} L &= \min\{0 \leq j \leq H - K + 1 : n_{j+\ell}^+ \geq r \forall 0 \leq \ell < K\}, \\ U &= \min\{0 \leq j \leq H - K + 1 : n_{j+\ell}^- \geq r \forall 0 \leq \ell < K\}. \end{aligned} \quad (27)$$

Whenever both sets are nonempty, $T_{r,K}$ is finite and $L \leq T_{r,K} \leq U$. No monotonicity of accuracy is assumed.

GREENCERT in six steps. At an eligible checkpoint: (1) build the anchor-affine HVP clock and apply the predeclared variational sweeps; (2) freeze the first predicted future event and its horizon; (3) compute $s, z = K_H s$, $Z = \|z\|_2$, and $p = \|z\|_{\infty,2}$; (4) build neural jets on the prespecified outer domain ρ and apply deterministic response-aware abstention; (5) advance the output and Green Gram powers until the response-centered quadratic closes or the frozen ladder is exhausted; and (6) transport $\varrho = p + E$ to strict margin paths and return $[L, U]$ if both endpoints exist. The frozen cohorts use this step; Corollary 3 gives a tighter future replacement. Selection never uses a randomized probe or a future true trajectory. The fixed-radius studies use the matched specialization $\varrho = R = 2Z$.

4 Frozen studies

Protocol terminology. An *eligible* checkpoint satisfies the prespecified trigger condition, and a *trigger-only anchor* is chosen using trigger data without reading its future certification trajectory. A *candidate* is the resulting checkpoint–event pair. *Freezing* fixes a methodological choice before the relevant future outcome is inspected; *sealing* records the method, candidate, or certificate before the outcome join. An *issued* candidate satisfies every closure and event-margin condition; otherwise GREENCERT abstains. Post-seal analyses are labeled separately and never alter prospective issuance.

We run four outcome-sealed cohorts on three task families; complete splits, hyperparameters, selectors, and seal provenance appear in Appendix D. WDBC uses 24 fresh seeds, disjoint train/trigger/certification splits, a 266-parameter tanh MLP, and three gates (72 cases). Digits uses 12 fresh seeds, parity-stratified disjoint splits, a 538-parameter tanh MLP, and two gates (24 cases). Their selectors cannot read future certification trajectories. WDBC fixes $H = 300$ and four variational sweeps; digits fixes $H = 400$ and three. Both use a 10^{-6} family budget, seal candidate and certificate records before outcome joins, and receive an independent 192-bit zero-radius continuation audit.

Two separately sealed, disjoint-seed Transformer cohorts use the same one-block, 13,792-parameter causal model on addition modulo 17, three gates, $K = 25$, $H = 300$, and four sweeps [Vaswani et al., 2017]. The first freezes the fixed-radius rule $R = 2Z$ at $m = 16, q = 8$; the second seals the response-centered root and shared-probe ladder $q = 1, \dots, 8$ before its seeds run. Future trajectories remain inaccessible through certificate sealing. The earlier fixed-radius selector reads only the current certification count, so we describe both cohorts as prospective/outcome-sealed rather than untouched. One sealed response-centered horizon mismatch is retained as a pre-outcome construction abstention (Appendix H).

Matched unsigned propagation, the 0–4-sweep mechanism study, batched probes, and the two-block LayerNorm/AdamW transport test are explicitly post-seal or development audits. Dense outward checks and a disjoint active-subspace study separate numerical rigor from the headline confirmations. No diagnostic audit changes a prospective issuance count.

5 Results

5.1 A non-modular, information-separated confirmation

The WDBC protocol freezes 71 candidates among 72 prespecified seed-gate cases; one case has no trigger-only anchor. Without inspecting a future certification trajectory, GREENCERT issues 56 singleton brackets across 22 of 24 seeds. After the sealed join, all 56 contain the first persistent crossing. Median lead is 79 optimizer updates and maximum lead is 256. The four-sweep clock is exactly timed in all 59 candidates with a comparable finite certification-set event: the 56 strictly future crossings plus three gates already present at the anchor. GREENCERT issues on the 56 strictly future cases. The 15 candidate abstentions comprise those three step-zero events and 12 cases with no strictly future centerline event.

Table 1: Headline prospective/frozen studies. Coverage is conditional on issuance; all denominators and matched directional comparisons are shown.

Study	Cases	Candidates	Issued	Covered	Matched comparison
WDBC	72	71	56	56/56	unsigned also issues 56
Digits	24	24	7	7/7	signed-only [147, 147]; 2/2 inaccurate clocks abstain
Transformer fixed- R	72	23	9	9/9	unsigned retains 1/9 signed certificates
Transformer response	72	19	11	11/11	fixed- R baseline issues 10

The 63 WDBC and digits brackets additionally receive independent exact-real continuation audits. The 20 Transformer certificates remain high-confidence numerical claims under the stated float64 and ideal-PRNG assumptions.

Thresholds within a seed are correlated, so we do not attach event-level binomial confidence intervals to conditional coverage. At the seed level, all issued events cover in 22/22 WDBC issuing seeds, 6/6 digits seeds, 6/6 fixed-radius Transformer seeds, and 7/7 response-centered Transformer seeds. These are implementation audits of a certificate, not estimates of a universal forecast-accuracy rate.

The result survives an independent 192-bit outward replay of 40 unique seed-anchor tubes and 5,089 optimizer transitions. All requested horizons are reached and all 56 brackets are retained unchanged. The maximum outward radius is 4.37×10^{-13} , minimum strict output slack is 2.38×10^{-5} , and the smallest recorded recurrence-rounding slack is 3.94×10^{-31} . This is a direct validated continuation, not a replay of the Green inequality: it begins with zero anchor radius and encloses each optimizer step, with no κ , Gaussian event, or PRNG dependence. Hence the retained brackets are deterministic exact-real statements for continuation from the stored dyadic checkpoint; the preceding PyTorch/BLAS training program and population accuracy are not certified.

The same direct pass takes 7.12 aggregate hours versus 33.88 minutes for matched GreenCert event records ($12.61 \times$). Direct outward continuation provides an independent validation baseline on the small tanh networks, but it does not provide the matrix-free signed-Green construction used for the Transformer setting. GREENCERT separates the realized first-order response from the nonlinear remainder and accesses the finite-window gain through JVP/VJP products, avoiding explicit propagator or Hessian construction. The direct 192-bit runs therefore audit the issued small-model brackets rather than replace the matrix-free certification mechanism.

On this locally near-affine tanh regime the matched unsigned finite-window baseline also issues 56/56. Directional separation is evaluated in the prospective digits study and Transformer audit below.

5.2 A prospective non-modular signed separation

All 24 prespecified digits seed-gate cases yield trigger-only anchors. Before any future certification trajectory is opened, the signed method issues seven certificates across six seeds; after the sealed join, all seven contain the event. Leads are 42–147 updates (median 70), and all seven brackets are singletons. The matched unsigned method issues six. The predeclared signed-only case is seed 509’s 90% event: from anchor 35, the first persistent crossing occurs at relative step 147, exactly the frozen bracket [147, 147]. Its signed closure statistic is 0.2369, while discarding direction raises the zero-order response by $54.94\times$ and the unsigned closure to 13.018. Minimum strict output slack is 1.43×10^{-5} . Twelve operators were queried, giving a realized family bound of 5×10^{-7} .

The frozen centerline has a finite prediction and revealed event in 16 cases. Fourteen offsets are exact. The remaining two are seed 510 at 92.5% (prediction 218, event 219) and seed 511 at 90% (prediction 358, event 361). Both are outcome-blind safe early abstentions: even the favorable $\kappa \geq 1$ closure lower bounds are 6,512 and 4,505, respectively, so neither case consumes a randomized Green query. Thus the prospective study includes both sides of selective validity: every issued bracket covers, while every inaccurate finite centerline is rejected.

A Green-independent 192-bit replay then encloses seven exact-real seed-anchor tubes from zero radius. All seven horizons are reached and all seven singleton brackets are retained unchanged, including [147, 147]; the maximum outward radius is 5.07×10^{-9} and minimum strict logic slack is 1.46×10^{-5} . This validates continuation from stored dyadic checkpoints, not the preceding training execution or the randomized Green inequality.

This is a prospective, non-modular instance in which direction changes the verifier’s decision, not merely its radius. Appendix F records all issued cases and the seal chain.

5.3 Prospective fixed-radius Transformer confirmation

Across 72 prespecified seed-gate cases, 19 never become eligible, one already has its gate at the first admissible point, and 29 have no future clock event in the causal scan. The remaining 23 candidates span 12 distinct seeds. Before any future outcome is opened, GREENCERT issues nine certificates across six seeds. After the sealed join, all nine contain the first 25-update persistent crossing. Every bracket is a singleton; leads range from 28 to 274 updates, with median 192. Issuance is 9/72 (12.5%) over all prespecified cases and 9/23 (39.1%) conditional on a frozen candidate.

The frozen four-sweep centerline predicts the event offset exactly in all 23 candidates. Certification is stricter: only nine candidates satisfy both nonlinear closure and output-margin conditions, separating empirically accurate forecasts from forecasts for which the realized nonlinear trajectory is formally enclosed. Of the 14 abstentions, one is a retained construction failure, four fail the safe pre-probe lower bound, and nine fail the probed Green closure.

The numerical boundaries are not close. Minimum nonlinear closure slack is 0.1445, minimum strict output-logic slack is 2.67×10^{-5} , and the largest observed issued sequence-error/radius ratio is 0.5028. No issued sequence tube or pointwise state tube is violated after outcome reveal. The execution queried 4,961 randomized operators, for a realized union bound 2.28×10^{-7} under the frozen Gaussian model, below the declared 10^{-6} family ceiling.

Coverage is conditional on issuance: all certificates in each of six issuing seeds cover. Appendix G gives cluster-aware sensitivity summaries and every issued and abstained candidate.

5.4 Separately sealed response-centered Transformer confirmation

The fresh seeds 355–378 produce 19 frozen candidates across 11 seeds from 72 prespecified seed-gate cases; 42 cases have no future modal candidate and 11 never become trigger-eligible. Before any future trajectory is opened, the response-centered rule issues 11 certificates. After the certificate seal, all 11 contain the first persistent crossing, across seven distinct issuing seeds. Every bracket is a singleton; median lead is 118 updates and maximum lead is 275. The first issuing powers are one case at $q = 1$, seven at $q = 2$, two at $q = 3$, and one at $q = 4$; no issued case needs powers 5–8. The matched power-8 fixed-radius baseline issues 10. The additional case is seed 372’s 70% gate, where the frozen bracket [246, 246] equals the revealed crossing.

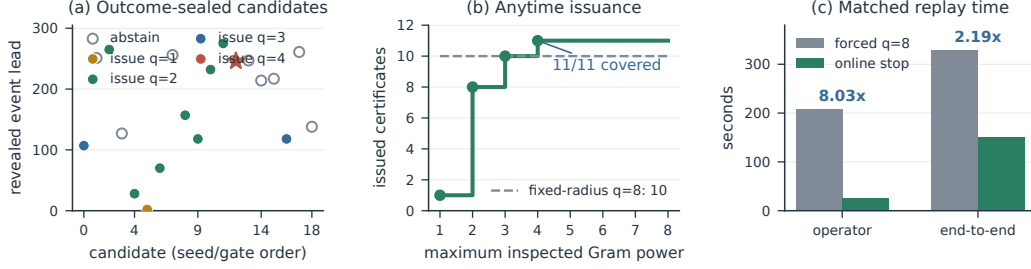


Figure 1: Separately sealed response-centered confirmation and executable anytime stopping. (a) All 19 frozen candidates; filled points issue and color gives the first valid power. The red star marks the one v3-only certificate over the fixed-radius baseline. (b) All 11 certificates are available by power four; the matched fixed-radius rule issues 10 at power eight. (c) On one sealed candidate, online and forced-q8 executions use identical code and streams and return the same bracket, while online stopping cuts operator and end-to-end time by $8.03\times$ and $2.19\times$.

All 19 finite modal offsets equal the revealed outcomes, so selective rejection of inaccurate clocks remains the role of the independently sealed digits study. No issued total-state tube is violated. The execution queries 3,649 randomized operators, giving realized family failure bound 1.68×10^{-7} under the committed Gaussian model.

The response-centered decomposition also exposes the Transformer’s numerical boundary more sharply. In float64, subtractive cancellation makes the raw observed $\|h - z\|_X$ diagnostic exceed its nominal second-order E in all 11 issued cases, by at most 3.07×10^{-14} in absolute radius. This does not alter any observed total-state containment or event bracket, and every issued bracket survives an added 10^{-8} radius floor (more than 3×10^5 times that deficit), but the padding study is sensitivity evidence rather than an outward rounding proof. We therefore report both Transformer cohorts as predeclared high-confidence numerical certificates, not computer-assisted exact-real proofs.

5.5 Signed defect propagation controls certificate issuance

The matched finite-window baseline makes the mechanism unusually sharp. Of the 18 frozen Transformer candidates for which the Green operator was queried, the signed construction issues nine certificates. The unsigned replacement $Z_u = \kappa \|s\|_2$ retains only one. Eight of nine signed certificates therefore disappear *solely* because defect direction is discarded. Across all 18 matched cases, Z_u/Z ranges from 83.0 to 411.5, with median 319.7. Seventeen cases are ruled out analytically even under the unsigned method’s favorable outer-ball envelope; the sole survivor retains the same $[30, 30]$ event bracket but needs a radius $60.6\times$ larger. The independently sealed digits study reaches the same conclusion away from modular arithmetic: signed propagation issues 7/24 cases versus 6/24 unsigned, with the additional 147-step event covered exactly. Thus the directional ablation now changes issuance in two datasets and two parameterizations, including one protocol-primary non-modular case.

On three burned candidates, four development-fixed sweeps reduce median maximum defect from 8.57×10^{-3} to 5.94×10^{-8} (1.44×10^5) and make timing errors $(-9, -1, -3)$ exact. Each costs H HVPs: the path is an a-posteriori pseudo-orbit solve, not a single-Hessian clock.

5.6 A directional fourth-order remainder removes one sweep

The cancellation-safe second response contains three known copies of the parameter correction and one unknown dual direction. Replacing all four by a single Euclidean operator norm was the dominant obstruction in a frozen three-sweep audit. Corollary 8 instead propagates a polarized block majorant and maximizes only over the free slot. Across 14 nondevelopment cases, the resulting Taylor-sequence bound is smaller at every checkpoint; its ratio to the scalar bound ranges from 9.09×10^{-5} to 1.02×10^{-3} , with median 3.45×10^{-4} ($2,899\times$ tighter).

Under the unchanged recorded Green bounds, three holdout closures that previously abstained now issue $[28, 28]$, $[70, 70]$, and $[118, 118]$; the development row retains $[2, 2]$. All four brackets exactly

match their sealed four-sweep comparators, and all strict event slacks exceed 1.36×10^{-3} . Thus an adaptive rule can stop after three predeclared centerline sweeps on four of 15 records, saving four of 60 centerline sweeps in the fixed panel without changing its prospective coverage count.

An independent bivariate mixed jet propagates only the coefficient of $t^3\varepsilon$ rather than materializing the degree-four polynomial. It reproduces every checkpoint bound and closure decision with maximum relative error 4.19×10^{-15} . Its median whole-case runtime improves $154.31 \rightarrow 115.92$ seconds ($1.33\times$); this misses the prespecified $2\times$ speed gate because centerline replay dominates, so we claim the equivalence and sweep saving, not a twofold end-to-end acceleration. The audit is post-release, outcome-blind, and makes no new randomized query.

5.7 Computational scaling with batched operator products

Post-seal batching accelerates complete replay by $3.00/2.60\times$. Anytime stopping exposes all 11 certificates by $q \leq 4$ (median two) and, in a matched replay, preserves $[2, 2]$ while cutting operator/end-to-end cost by $8.03 \times /2.19\times$.

Post-seal, the cancellation-safe response closes 15/15 Green-evaluable records versus 11/15, saving 22 powers and 167,250 net objective HVPs; its nine fourth-order gates retain at least $55.3\times$ headroom. At $\lambda = 4096$, the amplified secant reproduces $[28, 28]$ with $24.4\times$ analytic headroom. Its 204 outward 192-bit scalar jets bound forcing by 1.31×10^{-29} and retain $24.41\times$ headroom in 9.80 minutes wall time. Three rotating-order triples give 6.56 s versus 8.10 s for third-order AD and 26.01 s for the next power, a $4.04\times$ paired speedup (Appendix I).

Corrected-path relinearization removes the mixed coefficient, $0.627 \rightarrow 0$. In a frozen outcome-blind 15-case panel, every directional bracket survives; 14 stop at four probes and one at eight. Green applications fall $560 \rightarrow 64$ ($8.75\times$) and linearized sweeps $1150 \rightarrow 144$ ($7.99\times$). Causal prefix construction is bitwise identical and gives $9.94/2.09/1.06\times$ speedups at horizons 26/131/299; direct-image release skips K^T in 4/15 cases. These post-seal audits leave frozen prospective counts above unchanged.

Scaled-momentum structure gives a second, orthogonal reduction. Restricting the nonlinear gain to $\Pi_\theta K_H \mathcal{B}$ preserves all 15 brackets, converts two additional cases to direct-image release, and reduces the matched staged total from 112 to 96 logical Green sweeps. A stricter anchor-zero restriction preserves every bracket but remains at 96 sweeps; it therefore fails its prespecified promotion gate and is retained as a negative systems audit rather than advertised as another speedup.

The deterministic neural-jet release sharpens the practical picture further. Across the complete 15-case operator cohort it preserves eight brackets with no randomized output-Jacobian query, eliminating 1,432 output operators and 22,912 $q = 1$ Gram applications. All seven unresolved cases retain the existing probe fallback, so the staged policy still issues 15/15 with identical brackets. For the eight deterministic releases the failure upper is the 10^{-6} Green event alone; the minimum strict event-logic slack is 1.80×10^{-4} . An independent implementation reproduces every disposition and bracket without reading a future outcome.

The same ingredients were finally composed in one outcome-blind execution on the sealed seed-366 80% case ($H = 26$). Three separately launched runs retain $[2, 2]$ in 9.01/11.71/9.21 s (median 9.21 s), using four applications of K_H , no application of $K_H^\top$, and no randomized output-Jacobian operator. An independent audit reconstructs the analytic closure, event logic, hashes, and timing ratios from the committed records. The historical fixed- $q = 8$ cross-batch median was 58.02 minutes, a $378\times$ reduction in recorded construction time. On the same seed, anchor, optimizer, and machine, ordinary continuation takes 0.298 s for 26 updates and 3.718 s for 300; certification is therefore still $30.9\times$ and $2.48\times$ slower, respectively. These matched controls separate the practical gain from the different question of whether revealing the future is cheaper than proving it.

6 Related Work

Forecasting and grokking. Modular arithmetic is a standard grokking testbed [Power et al., 2022, Gromov, 2023, Mohamadi et al., 2024]; geometry and clocks predict transitions [Notsawo et al., 2023, Kim, 2026, Khanh et al., 2026], and local quadratic forecasts reach 150M parameters [Meterez et al., 2026]. GREENCERT instead certifies finite-window nonlinear validity and first passage.

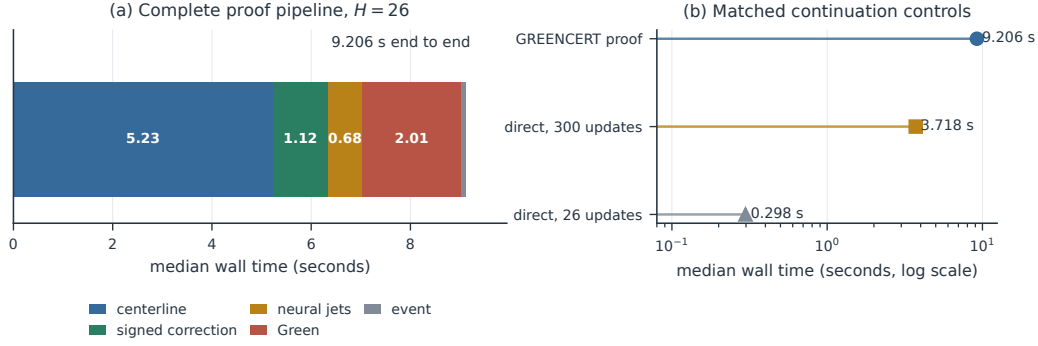


Figure 2: Measured cost of the fully composed verifier on the sealed seed-366, $H = 26$ case. (a) Median phase times over three separately launched executions; all retain $[2, 2]$. (b) Same-machine direct-continuation controls from the same seed, anchor, and optimizer. The 58.02-minute historical fixed- $q = 8$ cross-batch median is excluded from the matched panel.

Trajectory verification and robust training. Minute perturbations can send otherwise identical neural-training trajectories to different solutions [Altıntaş et al., 2025], making the realized checkpoint—rather than an interchangeable nearby orbit—the relevant anchor for an event claim. Classical shadowing, pseudo-orbit, and radii-polynomial methods certify trajectory proximity/existence, including in learning dynamics [Chow et al., 1989, Chow and Palmer, 1992, Chow and Vleck, 1994, Hammel et al., 1987, Sauer and Yorke, 1991, Lessard and Reinhardt, 2014, Bielecki and Ombach, 2004, Orvieto and Lucchi, 2019]. Product-space radii bounds can preserve component structure [van den Berg and Jaquette, 2018], while goal-oriented a posteriori methods use adjoint residuals to control selected quantities of interest [Becker and Rannacher, 2001]. Our forcing-subspace result uses the exact scaled-momentum factorization to close the anchored nonlinear recurrence in parameter space before transporting it through persistent neural-output margins. Robust training/parameter balls [Sosnin et al., 2026, Taheri et al., 2026, Weng et al., 2020], descent/stability [Streeter and Dillon, 2022, Shi et al., 2026], and fixed-network or surrogate verification [Wicker et al., 2021, Mathiesen et al., 2026, Naughton, 2026] deliver different guarantees. GRENCERT propagates signed endogenous defect on the anchored optimizer orbit to a persistent event bracket. Matrix-free tools [Pearlmutter, 1994, Kuczyński and Woźniakowski, 1992] and inexact-Newton residual interfaces [Dembo et al., 1982] make that causal certificate executable. The directional fourth-order construction also draws on classical polarization of symmetric multilinear forms [Thomas, 2014] and multivariate Taylor-model arithmetic [Berz and Hoffstätter, 1998]. Taylor models have also been integrated with neural-controller reachability [Schilling et al., 2022], while recent neural verification preserves derivative information inside local reachability bounds [Sharifi and Fazlyab, 2024, Entesari and Fazlyab, 2026]. Here the polynomial is evaluated on an endogenous optimizer correction and feeds a finite-window first-passage closure rather than an exogenous input set.

7 Limitations and broader impact

The complete certificates currently cover shallow tanh MLPs and a one-block, 13,792-parameter Transformer. A 102,400-parameter LayerNorm/AdamW audit transports the required derivatives but does not close the event theorem; deep networks and stochastic optimization remain open. At 1.01M parameters, the matrix-free core takes 11.37 hours before neural-jet construction. The fully composed 13,792-parameter $H = 26$ verifier now runs in 9.21 s median, but matched direct continuation is still $30.9\times$ faster over the needed window, and this small-model result does not resolve modern-scale closure. The events are defined on fixed evaluation sets and coverage is conditional on issuance, so these are not population guarantees.

The directional-block remainder is presently specialized to the same normalization-free Transformer and uses analytic block majorants rather than outward-rounded coefficient arithmetic. Its post-release audit establishes theorem tightness and sweep reduction on a frozen panel; it is not counted as an additional prospective coverage experiment.

Table 2: Objects and guarantees delivered by representative approaches.

Approach	State/object	Uncertainty or correction	Delivered claim
Chow–Lin–Palmer; Chow–Palmer [Chow et al., 1989, Chow and Palmer, 1992]	numerical pseudo-orbit	right-inverse/Newton correction	shadowing orbit and proximity bound
Lessard–Reinhardt [Lessard and Reinhardt, 2014]	IVP coefficient sequence	residual and radii polynomial	rigorous solution enclosure
AutoBound/SafeRate [Streeter and Dillon, 2022]	one optimizer step	Taylor-remainder majorizer	certified loss decrease
YES training bounds [Yeganegi et al., 2024]	layerwise data/output mappings	linear/intermediate baselines	online training-quality and loss bounds
Abstract Gradient Training [Sosnin et al., 2026]	reachable training parameters	training-data perturbation set	robust prediction after training
Barrier robustness [Taheri et al., 2026]	family of poisoned training trajectories	learned barrier and scenario bound	terminal poisoning-robustness radius
Adaptive SGD stopping [Aolaritei et al., 2026]	strongly convex stochastic trajectory	time-uniform confidence sequence	distance/suboptimality stopping certificate
CT-BaB [Shi et al., 2026]	neural controller and input region	verification-aware branch and bound	post-training Lyapunov stability
Certified neural dynamics [Wicker et al., 2021, Mathiesen et al., 2026]	learned/Bayesian dynamical model	model or transition uncertainty	reachability or surrogate-trajectory enclosure
Certified Grokking [Naughton, 2026]	fixed published checkpoint and circuit	outward intervals and Lean theorem	full-domain endpoint and circuit certificate
Weight-space robustness [Weng et al., 2020]	fixed trained network	exogenous parameter ball	robust prediction at that network
GREENCERT	optimizer orbit from realized anchor	signed temporal defect and nonlinear tube	persistent first-passage bracket or abstention

The 63 direct 192-bit proofs begin at stored checkpoints. The 204 amplified-secant jets are outward at 192 bits conditional on stored dyadic inputs, whereas the upstream Transformer Green products, derivative envelopes, and output margins remain float64/ideal-PRNG calculations. The post-seal $8.75\times$ Green reduction and eight deterministic output releases reuse frozen analytic envelopes. They reduce the present cost substantially, but they do not settle neural-jet construction at modern scale.

8 Conclusion

This work asks when a checkpoint-local training clock can support a statement about the realized future trajectory. GREENCERT answers selectively: it returns a persistent-event bracket when signed response and nonlinear closure can be verified, and abstains otherwise. Across the frozen studies, 83/83 issued brackets cover and both inaccurate finite clocks abstain. The final implementation preserves all 15 audited Transformer brackets while cutting Green work $560 \rightarrow 64$; deterministic neural jets remove every output probe in eight of those cases. A directional block remainder additionally recovers three holdout certificates under a three-sweep reference while an independent mixed jet reproduces every local bound. Their complete composition turns one sealed Transformer certificate from an hours-scale historical construction into a reproducible 9.21-second proof object without changing its bracket.

References

- Ethem Alpaydin and Cenk Kaynak. Optical recognition of handwritten digits. UCI Machine Learning Repository, 1998. CC BY 4.0; scikit-learn distributes a 1,797-example copy of the test subset.
- Gül Sena Altuntaş, Devin Kwok, Colin Raffel, and David Rolnick. The butterfly effect: Neural network training trajectories are highly sensitive to initial conditions. In *Proceedings of the*

- 42nd International Conference on Machine Learning, volume 267 of *Proceedings of Machine Learning Research*, pages 1314–1342, 2025. URL <https://proceedings.mlr.press/v267/altintas25a.html>.
- Liviu Aolaritei, Lucas Lévy, Francis Bach, and Michael I. Jordan. Beyond optimal rates in stochastic optimization: Trajectory-adaptive stopping rules. *arXiv preprint arXiv:2608.25551*, 2026. URL <https://arxiv.org/abs/2608.25551>.
- Roland Becker and Rolf Rannacher. An optimal control approach to a posteriori error estimation in finite element methods. *Acta Numerica*, 10:1–102, 2001. doi: 10.1017/S0962492901000010.
- Martin Berz and Georg Hoffstätter. Computation and application of taylor polynomials with interval remainder bounds. *Reliable Computing*, 4(1):83–97, 1998. doi: 10.1023/A:1009958918582.
- Andrzej Bielecki and Jerzy Ombach. Shadowing property in analysis of neural networks dynamics. *Journal of Computational and Applied Mathematics*, 164–165:107–115, 2004. doi: 10.1016/S0377-0427(03)00486-2.
- Daniel Carando and Jorge Tomás Rodríguez. Symmetric multilinear forms on Hilbert spaces: Where do they attain their norm? *Linear Algebra and its Applications*, 563:178–192, 2019. doi: 10.1016/j.laa.2018.10.023.
- Shui-Nee Chow and Kenneth J. Palmer. On the numerical computation of orbits of dynamical systems: The higher dimensional case. *Journal of Complexity*, 8(4):398–423, 1992. doi: 10.1016/0885-064X(92)90004-U.
- Shui-Nee Chow and Erik S. Van Vleck. A shadowing lemma approach to global error analysis for initial value ODEs. *SIAM Journal on Scientific Computing*, 15(4):959–976, 1994. doi: 10.1137/0915058.
- Shui-Nee Chow, Xiao-Biao Lin, and Kenneth J. Palmer. A shadowing lemma with applications to semilinear parabolic equations. *SIAM Journal on Mathematical Analysis*, 20(3):547–557, 1989. doi: 10.1137/0520038.
- Ron S. Dembo, Stanley C. Eisenstat, and Trond Steihaug. Inexact newton methods. *SIAM Journal on Numerical Analysis*, 19(2):400–408, 1982. doi: 10.1137/0719025.
- Taha Entesari and Mahyar Fazlyab. Hierarchical end-to-end taylor bounds for complete neural network verification. In *Proceedings of the 8th Annual Learning for Dynamics and Control Conference*, volume 331 of *Proceedings of Machine Learning Research*, pages 1070–1087, 2026. URL <https://proceedings.mlr.press/v331/entesari26a.html>.
- Andrey Gromov. Grokking modular arithmetic. *arXiv preprint arXiv:2301.02679*, 2023. URL <https://arxiv.org/abs/2301.02679>.
- Stephen M. Hammel, James A. Yorke, and Celso Grebogi. Do numerical orbits of chaotic dynamical processes represent true orbits? *Journal of Complexity*, 3(2):136–145, 1987. doi: 10.1016/0885-064X(87)90024-0.
- Truong Xuan Khanh, Truong Quynh Hoa, Luu Duc Trung, and Phan Thanh Duc. First-passage prediction of grokking delay: A calibrated law under AdamW with causal validation. *arXiv preprint arXiv:2605.18845*, 2026. URL <https://arxiv.org/abs/2605.18845>.
- Taeyoung Kim. Grokking on the weight-decay clock: A rate hierarchy from softly broken symmetries. *arXiv preprint arXiv:2607.23967*, 2026. URL <https://arxiv.org/abs/2607.23967>.
- Nikhil Kodali, Kartick Ramakrishnan, and Phani Motamarri. Residual-based Chebyshev filtered subspace iteration for sparse Hermitian eigenvalue problems tolerant to inexact matrix–vector products, 2025. URL <https://arxiv.org/abs/2503.22652>.
- Jacek Kuczyński and Henryk Woźniakowski. Estimating the largest eigenvalue by the power and lanczos algorithms with a random start. *SIAM Journal on Matrix Analysis and Applications*, 13(4): 1094–1122, 1992. doi: 10.1137/0613066.

- Jean-Philippe Lessard and Christian Reinhardt. Rigorous numerics for nonlinear differential equations using Chebyshev series. *SIAM Journal on Numerical Analysis*, 52(1):1–22, 2014. doi: 10.1137/13090883X.
- Frederik Baymler Mathiesen, Nikolaus Vertovec, Francesco Fabiano, Luca Laurenti, and Alessandro Abate. Certified neural approximations of nonlinear dynamics. In *AI Verification: Third International Symposium, SAIV 2026, Proceedings*, volume 16831 of *Lecture Notes in Computer Science*, pages 73–98. Springer, 2026. doi: 10.1007/978-3-032-32357-6_4. URL https://link.springer.com/chapter/10.1007/978-3-032-32357-6_4.
- Alexandru Meterez, Pranav Ajit Nair, Depen Morwani, Cengiz Pehlevan, Sham Kakade, and Alex Damian. A defense of the quadratic model. *arXiv preprint arXiv:2607.21716*, 2026. URL <https://arxiv.org/abs/2607.21716>.
- Mohamad Amin Mohamadi, Zhiyuan Li, Lei Wu, and Danica J. Sutherland. Why do you grok? a theoretical analysis on grokking modular addition. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 35934–35967, 2024. URL <https://proceedings.mlr.press/v235/mohamadi24a.html>.
- Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=9XFSbDPmdW>.
- Brian Naughton. Certified grokking: A machine-checked certificate for the nanda et al. modular-addition transformer, July 2026. URL <https://doi.org/10.5281/zenodo.21893060>. Exact-real full-domain checkpoint and circuit certificate.
- Pascal Junior Tikeng Notsawo, Hattie Zhou, Mohammad Pezeshki, Irina Rish, and Guillaume Dumas. Predicting grokking long before it happens: A look into the loss landscape of models which grok. *arXiv preprint arXiv:2306.13253*, 2023. URL <https://arxiv.org/abs/2306.13253>.
- Antonio Orvieto and Aurelien Lucchi. Shadowing properties of optimization algorithms. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Barak A. Pearlmutter. Fast exact multiplication by the hessian. *Neural Computation*, 6(1):147–160, 1994. doi: 10.1162/neco.1994.6.1.147.
- Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022. URL <https://arxiv.org/abs/2201.02177>.
- Tim Sauer and James A. Yorke. Rigorous verification of trajectories for the computer simulation of dynamical systems. *Nonlinearity*, 4(3):961–979, 1991. doi: 10.1088/0951-7715/4/3/018.
- Christian Schilling, Marcelo Forets, and Sebastián Guadalupe. Verification of neural-network control systems by integrating taylor models and zonotopes. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 8169–8177, 2022. doi: 10.1609/aaai.v36i7.20790. URL <https://ojs.aaai.org/index.php/AAAI/article/view/20790>.
- Sina Sharifi and Mahyar Fazlyab. Provable bounds on the hessian of neural networks: Derivative-preserving reachability analysis. *arXiv preprint arXiv:2406.04476*, 2024. URL <https://arxiv.org/abs/2406.04476>.
- Zhouxing Shi, Haoyu Li, Cho-Jui Hsieh, and Huan Zhang. Certified training with branch-and-bound for lyapunov-stable neural control. In *Proceedings of the 8th Annual Learning for Dynamics and Control Conference*, volume 331 of *Proceedings of Machine Learning Research*, pages 788–808, 2026. URL <https://proceedings.mlr.press/v331/shi26a.html>.
- Valeria Simoncini. Variable accuracy of matrix–vector products in projection methods for eigencomputation. *SIAM Journal on Numerical Analysis*, 43(3):1155–1174, 2005. doi: 10.1137/040605333.
- Valeria Simoncini and Daniel B. Szyld. Theory of inexact Krylov subspace methods and applications to scientific computing. *SIAM Journal on Scientific Computing*, 25(2):454–477, 2003. doi: 10.1137/S1064827502406415.

- Philip Sosnin, Matthew Wicker, Josh Collyer, and Calvin Tsay. Abstract gradient training: A unified certification framework for data poisoning, unlearning, and differential privacy. *Journal of Machine Learning Research*, 27(128):1–58, 2026. URL <https://www.jmlr.org/papers/v27/25-2206.html>.
- W. Nick Street, William H. Wolberg, and Olvi L. Mangasarian. Nuclear feature extraction for breast tumor diagnosis. In *Biomedical Image Processing and Biomedical Visualization*, volume 1905, pages 861–870. SPIE, 1993. doi: 10.1117/12.148698.
- Matthew Streeter and Joshua V. Dillon. Automatically bounding the taylor remainder series: Tighter bounds and new applications. *arXiv preprint arXiv:2212.11429*, 2022. URL <https://arxiv.org/abs/2212.11429>.
- Sara Taheri, Mahalakshmi Sabanayagam, Debarghya Ghoshdastidar, and Majid Zamani. Robustness certificates for neural networks against data poisoning and evasion attacks. *IEEE Open Journal of Control Systems*, 5:303–318, 2026. doi: 10.1109/OJCSYS.2026.3689712. URL <https://arxiv.org/abs/2512.20865>.
- Erik G. F. Thomas. A polarization identity for multilinear maps. *Indagationes Mathematicae*, 25(3): 468–474, 2014. doi: 10.1016/j.indag.2013.11.003. URL <https://arxiv.org/abs/1309.1275>.
- Jan Bouwe van den Berg and Jonathan Jaquette. A proof of wright’s conjecture. *Journal of Differential Equations*, 264(12):7412–7462, 2018. doi: 10.1016/j.jde.2018.02.018.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL <https://proceedings.neurips.cc/paper/7181-attention-is-all-you-need>.
- Tsui-Wei Weng, Pu Zhao, Sijia Liu, Pin-Yu Chen, Xue Lin, and Luca Daniel. Towards certificated model robustness against weight perturbations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 6356–6363, 2020. doi: 10.1609/aaai.v34i04.6105.
- Matthew Wicker, Luca Laurenti, Andrea Patane, Nicola Paoletti, Alessandro Abate, and Marta Kwiatkowska. Certification of iterative predictions in Bayesian neural networks. In *Proceedings of the Thirty-Seventh Conference on Uncertainty in Artificial Intelligence*, volume 161 of *Proceedings of Machine Learning Research*, pages 1713–1723, 2021. URL <https://proceedings.mlr.press/v161/wicker21a.html>.
- William H. Wolberg, Olvi L. Mangasarian, and W. Nick Street. Breast cancer wisconsin (diagnostic). UCI Machine Learning Repository, 1993. CC BY 4.0.
- Farhang Yeganegi, Arian Eamaz, and Mojtaba Soltanalian. Data-aware training quality monitoring and certification for reliable deep learning. *arXiv preprint arXiv:2410.10984*, 2024. URL <https://arxiv.org/abs/2410.10984>.

A Proofs

Proof of Theorem 1. Put $e = h - \tilde{z}$. Equation (9) gives

$$e = (K_H s - \tilde{z}) + K_H N(\tilde{z} + e). \quad (28)$$

The anchor terms vanish because $h_0 = \tilde{z}_0 = 0$. For every remaining transition input, the integral Taylor identity gives

$$\|N_j(\tilde{z}_j + e_j) - N_j(\tilde{z}_j)\|_2 \leq M \|\tilde{z}_j\|_2 \|e_j\|_2 + \frac{1}{2} M \|e_j\|_2^2. \quad (29)$$

Since $\|(\|e_j\|_2^2)_j\|_2 \leq \|e\|_X^2$, we obtain

$$\|N(\tilde{z} + e) - N(\tilde{z})\|_U \leq M p \|e\|_X + \frac{1}{2} M \|e\|_X^2. \quad (30)$$

The first term in the fixed-point map is bounded by α , and $\|K_H N(\tilde{z})\|_X \leq \beta$. Thus Eq. (12) makes the continuous exact fixed-point map send the finite-dimensional closed radius- E sequence ball into itself. The domain condition validates every derivative bound. Brouwer gives a fixed point; causal forward recurrence makes it unique and equal to the realized trajectory error. $\square$

Proof of Corollary 1. Taylor's theorem yields

$$\|q\|_U \leq \frac{1}{2}M \left(\sum_{j=1}^{H-1} \|\tilde{z}_j\|_2^4 \right)^{1/2} \leq \frac{1}{2}MpZ. \quad (31)$$

Hence $\beta = \kappa MpZ/2$ is valid. Moving the right side of Eq. (12) to the left gives a convex quadratic in E with linear coefficient $bp - 1$ and constant Y . Conditions (15) make its discriminant nonnegative and select the smaller root; the displayed form is obtained by rationalizing the usual quadratic formula. $\square$

Proof of Corollary 2. The checkpointwise Taylor bounds give

$$\|N(\tilde{z})\|_U \leq Q, \quad (32)$$

$$\|N(\tilde{z} + e) - N(\tilde{z})\|_U \leq A\|e\|_X + \frac{1}{2}B\|e\|_X^2. \quad (33)$$

Applying K_H and repeating the self-mapping argument proves Eq. (18). Finally, $Q \leq MpZ/2$, $A \leq Mp$, and $B = M$ show dominance. $\square$

Proof of Corollary 3. Theorem 1 gives $\|x_{a+j} - \bar{c}_j\| \leq E$. Integrating the logit Hessian from c_j to $\bar{c}_j$ yields $\|Df(\bar{c}_j)\| \leq B_{1j} + B_{2j}d_j$. A second Taylor expansion from $\bar{c}_j$ to x_{a+j} , followed by the $\sqrt{2}$ norm of a difference of two logit coordinates, proves Eq. (26). Finally, $d_j \leq p$ and direct subtraction from Eq. (25) at $\varrho = p + E$ leave the nonnegative quantity $\sqrt{2}(B_{1j}p + B_{2j}p^2/2)$. $\square$

Proof of Proposition 1. Let v be a unit top eigenvector of $A = T^\top T$. For each probe,

$$\|A^q g_i\| \geq \|A\|^q |v^\top g_i|. \quad (34)$$

The projections are independent standard normals. By the definition of c_δ , the probability that all m absolute projections are below c_δ is $(2\Phi(c_\delta) - 1)^m = \delta$. Outside that event, $\|A\|^q \leq Y_q/c_\delta$ for every q on the same event. Since $\|A\| = \|T\|^2$, Eq. (20) follows simultaneously over all inspected powers. $\square$

Proposition 3 (Residual-corrected inexact Gram powers). *Let $A = T^\top T$ and fix an integer $q \geq 1$. Suppose computed iterates $\tilde{v}_{i,0} = g_i$ satisfy $\max_i \|\tilde{v}_{i,\ell+1} - A\tilde{v}_{i,\ell}\| \leq \xi_\ell$ for $0 \leq \ell < q$, and put $\tilde{Y}_q = \max_i \|\tilde{v}_{i,q}\|$. Let R_q be the unique positive root (or zero when all data vanish) of*

$$c_\delta R_q^q = \tilde{Y}_q + \sum_{\ell=0}^{q-1} \xi_\ell R_q^{q-1-\ell}. \quad (35)$$

Then, on the same probability- $1 - \delta$ event as Proposition 1, simultaneously over inspected powers, $\|T\| \leq \sqrt{R_q}$. The analogous Frobenius-block statement replaces $\tilde{Y}_q$ and ξ_ℓ by block Frobenius norms and c_δ by the lower δ -quantile of χ_m .

Proof of Proposition 3. Let $\lambda = \|A\| = \|T\|^2$ and let v be a unit top eigenvector. On the single Gaussian event from Proposition 1, some probe g_i satisfies $|v^\top g_i| \geq c_\delta$. With $r_{i,\ell} = \tilde{v}_{i,\ell+1} - A\tilde{v}_{i,\ell}$, telescoping gives

$$A^q g_i = \tilde{v}_{i,q} - \sum_{\ell=0}^{q-1} A^{q-1-\ell} r_{i,\ell}. \quad (36)$$

Consequently,

$$c_\delta \lambda^q \leq \tilde{Y}_q + \sum_{\ell=0}^{q-1} \xi_\ell \lambda^{q-1-\ell}. \quad (37)$$

The polynomial obtained from Eq. (35) has one positive leading coefficient and only nonpositive lower coefficients. Unless every datum vanishes, Descartes' rule gives exactly one positive root (zero may also be a root when trailing coefficients vanish). The displayed inequality thus forces $\lambda \leq R_q$. The projection event is common to every power.

For the block analogue, collect probes as $G = [g_1 \cdots g_m]$ and use $c_{\delta,m}^X = F_{\chi_m}^{-1}(\delta)$. Since $\|v^\top G\| \sim \chi_m$ and $\|A^q G\|_F \geq \lambda^q \|v^\top G\|$, the same proof applies with Frobenius terminal and residual norms. Splitting δ between the max-column and chi-block events makes the minimum of their two bounds valid by a union bound, without another Gram application. $\square$

Corollary 4 (Precision-adaptive Gram queries). *Fix the operator and Gaussian block in Proposition 3. The arithmetic precision, implementation, and residual tolerances at each power may be selected or refined after inspecting all preceding iterates. If the reported ξ_ℓ are valid upper bounds relative to the same exact A , the resulting R_q remains a simultaneous probability- $1 - \delta$ enclosure for every inspected power; precision refinement consumes no additional failure budget.*

Proof. On the single top-eigenvector projection event, the telescoping inequality in the preceding proof is deterministic and holds for every sequence of computed iterates whose displayed residual bounds are valid. Adaptive arithmetic therefore changes only the coefficients of the scalar polynomial, not the Gaussian event. $\square$

Corollary 5 (Certificate-aware operator-cap budget). *Fix $\bar{R} > 0$ and nonnegative $\bar{Y}, b_0, \dots, b_{q-1}$ with*

$$\bar{Y} + \sum_{\ell=0}^{q-1} b_\ell \leq c_\delta \bar{R}^q. \quad (38)$$

If $\tilde{Y}_q \leq \bar{Y}$ and $\xi_\ell \bar{R}^{q-1-\ell} \leq b_\ell$ for every $0 \leq \ell < q$, then, on the same probability- $1 - \delta$ event, $\|T\| \leq \sqrt{\bar{R}}$. Each failed local check may be recomputed at higher precision without another probe draw.

Proof. The assumptions make $\bar{R}$ a supersolution of Eq. (35). Proposition 3 and Corollary 4 complete the claim. $\square$

Inexact Krylov and variable-accuracy eigensolver theory likewise study perturbed matrix–vector products [Simoncini and Szyld, 2003, Simoncini, 2005]; recent residual-based Chebyshev iteration demonstrates the mixed-precision opportunity at large scale [Kodali et al., 2025]. Proposition 3 specializes residual transport to the one-sided randomized norm enclosure required here.

Proof of Proposition 2. Let E_n be the single-block event from Proposition 1 for operator T_n . Conditional independence gives $\Pr(E_n^c \mid \mathcal{F}_{n-1}) \leq \delta_n$. Therefore

$$\Pr\left(\bigcup_{n \geq 1} E_n^c\right) \leq \sum_{n \geq 1} \mathbb{E}[\Pr(E_n^c \mid \mathcal{F}_{n-1})] \leq \mathbb{E} \sum_{n \geq 1} \delta_n \leq \delta. \quad (39)$$

Every power and stopping decision for query n is already valid on E_n . $\square$

Quadratic defect contraction, Eq. (4). After one sweep, the new path defect is exactly

$$r_j^{(k+1)} = G(c_j^{(k)} + z_j^{(k)}) - G(c_j^{(k)}) - DG(c_j^{(k)})z_j^{(k)}. \quad (40)$$

The integral Taylor bound gives $\|r_j^{(k+1)}\|_2 \leq M \|z_j^{(k)}\|_2^2 / 2$. Taking maxima and using $\max_j \|z_j^{(k)}\|_2 \leq \kappa_{\infty,H} \max_i \|r_i^{(k)}\|_2$ proves the claim. $\square$

Proposition 4 (Inexact variational-sweep residual). *For a reference path c , let $r_j = G(c_j) - c_{j+1}$ and $J_j = DG(c_j)$. Let $\tilde{r}_j$ be a computed defect and let an anchor-fixed correction $\tilde{z}_0 = 0$ have recurrence residual*

$$d_j = \tilde{z}_{j+1} - J_j \tilde{z}_j - \tilde{r}_j. \quad (41)$$

The defect r'_j of the recentered path $c' = c + \tilde{z}$ satisfies the exact identity

$$r'_j = N_j(\tilde{z}_j) + (r_j - \tilde{r}_j) - d_j. \quad (42)$$

Consequently, if DG is M_j -Lipschitz on the connecting segment, $\|r_j - \tilde{r}_j\| \leq \sigma_j$, and $\|d_j\| \leq \tau_j$, then

$$\|r'_j\| \leq \frac{1}{2} M_j \|\tilde{z}_j\|^2 + \sigma_j + \tau_j. \quad (43)$$

Proof. Substitute the definition of r_j and add and subtract $J_j \tilde{z}_j$:

$$r'_j = G(c_j + \tilde{z}_j) - c_{j+1} - \tilde{z}_{j+1} \quad (44)$$

$$= N_j(\tilde{z}_j) + r_j + J_j \tilde{z}_j - \tilde{z}_{j+1}, \quad (45)$$

which is Eq. (42). Taylor's integral remainder and the triangle inequality give Eq. (43). $\square$

The residual term parallels classical inexact Newton forcing conditions [Dembo et al., 1982]; here it is resolved checkpointwise on the anchor-fixed causal sequence and carried into a first-passage verifier.

Corollary 6 (Residualized two-response interface). *Let $\hat{\kappa} \geq \|K_H\|$. Suppose $\tilde{s}$ and an anchor-fixed $\tilde{z}$ obey*

$$\|s - \tilde{s}\|_U \leq \sigma_s, \quad \|d^z\|_U \leq \tau_z, \quad d^z_j = \tilde{z}_{j+1} - J_j \tilde{z}_j - \tilde{s}_j. \quad (46)$$

Set $q_j = N_j(\tilde{z}_j)$. If a computed $\tilde{q}$ and anchor-fixed $\tilde{y}$ obey

$$\|q - \tilde{q}\|_U \leq \sigma_q, \quad \|d^q\|_U \leq \tau_q, \quad d^q_j = \tilde{y}_{j+1} - J_j \tilde{y}_j - \tilde{q}_j, \quad (47)$$

then Theorem 1 is valid with

$$\alpha = \hat{\kappa}(\sigma_s + \tau_z), \quad \beta = \|\tilde{y}\|_X + \hat{\kappa}(\sigma_q + \tau_q). \quad (48)$$

In particular, Proposition 3 may supply $\hat{\kappa}$ from residual-certified low-precision Gram products.

Proof. The recurrence definitions give the exact identities $\tilde{z} = K_H(\tilde{s} + d^z)$ and $\tilde{y} = K_H(\tilde{q} + d^q)$. Hence

$$\|K_H s - \tilde{z}\|_X \leq \hat{\kappa}(\sigma_s + \tau_z), \quad (49)$$

$$\|K_H q\|_X \leq \|\tilde{y}\|_X + \hat{\kappa}(\sigma_q + \tau_q). \quad (50)$$

These are precisely the α and β hypotheses of Theorem 1. $\square$

Corollary 7 (Cancellation-safe directional second response). *In the setting of Corollary 6, suppose G_j is C^3 on $c_j + t\tilde{z}_j$, $0 \leq t \leq 1$, and*

$$\sup_{0 \leq t \leq 1} \|D^3 G_j(c_j + t\tilde{z}_j)\| \leq L_j. \quad (51)$$

Define

$$q_j^{(2)} = \frac{1}{2} D^2 G_j(c_j)[\tilde{z}_j, \tilde{z}_j], \quad \sigma_2 = \left(\sum_{j=0}^{H-1} \left(\frac{1}{6} L_j \|\tilde{z}_j\|^3 \right)^2 \right)^{1/2}. \quad (52)$$

If $\|\tilde{q}^{(2)} - q^{(2)}\|_U \leq \sigma_{\text{ar}}$ and an anchor-fixed $\tilde{y}$ has recurrence residual

$$d^y_j = \tilde{y}_{j+1} - J_j \tilde{y}_j - \tilde{q}_j^{(2)}, \quad \|d^y\|_U \leq \tau_y, \quad (53)$$

then Theorem 1 may use

$$\boxed{\beta = \|\tilde{y}\|_X + \hat{\kappa}(\sigma_2 + \sigma_{\text{ar}} + \tau_y)}. \quad (54)$$

For scaled momentum $G(\theta, w) = (\theta - r, r)$, $r = \mu w + \eta \nabla F(\theta)$, and $\tilde{z}_j = (a_j, b_j)$,

$$q_j^{(2)} = \frac{\eta}{2} \left(-D^3 F(\theta_j)[a_j, a_j, \cdot], D^3 F(\theta_j)[a_j, a_j, \cdot] \right). \quad (55)$$

Consequently, $\|D^4 F\| \leq L_{F,4,j}$ on the parameter segment permits $L_j = \sqrt{2} \eta L_{F,4,j}$.

Proof. Third-order Taylor expansion gives $q_j = q_j^{(2)} + r_j^{(3)}$ with $\|r_j^{(3)}\| \leq L_j \|\tilde{z}_j\|^3/6$, hence $\|q - q^{(2)}\|_U \leq \sigma_2$. The recurrence identity $\tilde{y} = K_H(\tilde{q}^{(2)} + d^y)$ then yields

$$\|K_H q\|_X \leq \|\tilde{y}\|_X + \hat{\kappa}(\sigma_2 + \sigma_{\text{ar}} + \tau_y). \quad (56)$$

For scaled momentum, only the gradient term is nonlinear; differentiating it twice and three times gives Eqs. (55) and the stated fourth-objective bound. $\square$

Corollary 8 (Directional block remainder). *In Corollary 7, decompose parameter space as the orthogonal sum $\Theta = \bigoplus_{b=1}^B \Theta_b$, with projections π_b . For checkpoint j , let a_j be the parameter component of $\tilde{z}_j$ and $r_{j,b} = \|\pi_b a_j\|_2$. Suppose that on $\theta_j + ta_j$, $0 \leq t \leq 1$, the symmetric fourth derivative of F admits a nonnegative symmetric four-linear polarized block majorant $\mathcal{P}_{j,4}$:*

$$|D^4 F(\theta_j + ta_j)[h_1, h_2, h_3, h_4]| \leq \mathcal{P}_{j,4}(s(h_1), s(h_2), s(h_3), s(h_4)), \quad s(h)_b = \|\pi_b h\|_2. \quad (57)$$

This symmetry entails no loss: average any valid nonnegative four-linear majorant over all $4!$ slot permutations. Symmetry of $D^4 F$ preserves the inequality, and diagonal evaluation is unchanged. Write $\mathcal{P}_{j,4}(s) = \mathcal{P}_{j,4}(s, s, s, s)$. Then the scalar Taylor term in Eq. (52) may be replaced by

$$\sigma_{\text{blk}} = \left\{ \sum_{j=0}^{H-1} \left(\frac{\sqrt{2}\eta}{24} \|\nabla \mathcal{P}_{j,4}(r_j)\|_2 \right)^2 \right\}^{1/2}. \quad (58)$$

Consequently Eq. (54) remains valid with σ_2 replaced by σ_{blk} .

Proof. For a unit dual direction u , put $t_b = \|\pi_b u\|_2$; orthogonality gives $\|t\|_2 = 1$. Symmetry and homogeneity imply

$$\mathcal{P}_{j,4}(r_j, r_j, r_j, t) = \frac{1}{4} \nabla \mathcal{P}_{j,4}(r_j)^\top t \leq \frac{1}{4} \|\nabla \mathcal{P}_{j,4}(r_j)\|_2. \quad (59)$$

Taylor's integral remainder for the objective gradient contributes another factor $1/6$. The scaled-momentum map places the same parameter remainder in two state blocks with opposite signs, contributing $\sqrt{2}\eta$. Taking the Euclidean sequence norm gives Eq. (58). $\square$

The implementation obtains $\mathcal{P}_{j,4}$ coefficientwise through nonnegative sum, product, affine-map, and smooth-composition rules. A linear-cost alternative propagates only D_t^k and $D_t^k D_\varepsilon$ for $k \leq 3$; the terminal $D_t^3 D_\varepsilon$ coefficient is exactly the vector $\nabla \mathcal{P}_{j,4}(r_j)/4$ and avoids materializing degree-four monomials.

Corollary 9 (Amplified-secant second response). *In the setting of Corollary 6, choose $\lambda_j > 0$ and suppose G_j is C^3 on $c_j + t\tilde{z}_j$ for $0 \leq t \leq \max\{1, \lambda_j\}$, with $\|D^3 G_j\| \leq L_j$ there. Define*

$$q_j^{[\lambda_j]} = \frac{G_j(c_j + \lambda_j \tilde{z}_j) - G_j(c_j) - \lambda_j J_j \tilde{z}_j}{\lambda_j^2}, \quad \sigma_{\text{sec}} = \left(\sum_{j=0}^{H-1} \left[\frac{|\lambda_j - 1|}{6} L_j \|\tilde{z}_j\|^3 \right]^2 \right)^{1/2}. \quad (60)$$

If $\|\tilde{q}^{[\lambda]} - q^{[\lambda]}\|_U \leq \sigma_{\text{ar}}$ and an anchor-fixed $\tilde{y}$ has recurrence residual at most τ_y , then Theorem 1 may use

$$\boxed{\beta = \|\tilde{y}\|_X + \hat{\kappa}(\sigma_{\text{sec}} + \sigma_{\text{ar}} + \tau_y)}. \quad (61)$$

If the unscaled secant numerator has arithmetic error at most ε_j , checkpoint j contributes at most $A_j |\lambda_j - 1| + \varepsilon_j / \lambda_j^2$, where $A_j = L_j \|\tilde{z}_j\|^3 / 6$. For a λ -independent ε_j and $A_j, \varepsilon_j > 0$, the unconstrained optimum over $\lambda_j \geq 1$ is

$$\lambda_j^* = \max \left\{ 1, (2\varepsilon_j / A_j)^{1/3} \right\}, \quad (62)$$

clipped by the derivative domain and error budget. If $\varepsilon_j = 0$, take $\lambda_j = 1$; if $A_j = 0$, take the largest admissible amplification.

Proof. Taylor's integral identity gives

$$\frac{N_j(t\tilde{z}_j)}{t^2} = \int_0^1 (1-s) D^2 G_j(c_j + st\tilde{z}_j)[\tilde{z}_j, \tilde{z}_j] ds. \quad (63)$$

Subtracting the identities at $t = 1$ and $t = \lambda_j$, then integrating the $D^3 G_j$ drift, gives $\|N_j(\tilde{z}_j) - q_j^{[\lambda_j]}\| \leq |\lambda_j - 1| L_j \|\tilde{z}_j\|^3 / 6$. Equation (61) follows from the residualized two-response identity. Finally, differentiating $A_j(\lambda_j - 1) + \varepsilon_j / \lambda_j^2$ gives Eq. (62). $\square$

Corollary 10 (Fresh-probe forcing and Green-image residual). *Let $e \in \mathbb{R}^D$ be fixed before fresh independent $g_i \sim \mathcal{N}(0, I_D)$, $1 \leq i \leq m$. Suppose outward scalar intervals $[\ell_i, u_i]$ contain $g_i^\top e$, and define*

$$B = \max_i \max\{|\ell_i|, |u_i|\}, \quad c_\delta = \Phi^{-1}\left(\frac{1 + \delta^{1/m}}{2}\right). \quad (64)$$

Then, with conditional probability at least $1 - \delta$,

$$\boxed{\|e\|_2 \leq B/c_\delta}. \quad (65)$$

More sharply, for fixed $K : U \rightarrow X$, fresh $h_i \sim \mathcal{N}(0, I_X)$, and outward intervals containing $\langle K^\top h_i, e \rangle_U = \langle h_i, Ke \rangle_X$, the same construction gives $\|Ke\|_X \leq B/c_\delta$. Hence, with $e_{\text{ar}} = q^{[\lambda]} - \tilde{q}^{[\lambda]} - d^y$, one may use

$$\beta = \|\tilde{y}\|_X + \hat{\kappa}\sigma_{\text{sec}} + B/c_\delta, \quad (66)$$

without multiplying numerical error by $\hat{\kappa}$. Alternatively, probing $q^{[\lambda]}$ itself gives the response-free interface $\beta = \hat{\kappa}(\sigma_{\text{sec}} + B/c_\delta)$.

For scaled momentum, let a forcing probe at checkpoint j be $g_j = (g_{\theta,j}, g_{w,j})$ and put $w_j = g_{w,j} - g_{\theta,j}$. Its checkpoint contribution is the scalar bivariate jet

$$g_j^\top q_j^{[\lambda]} = \frac{\eta}{\lambda^2} \left[D_{w_j} F(\theta_j + \lambda a_j) - D_{w_j} F(\theta_j) - \lambda D_{w_j, a_j}^2 F(\theta_j) \right]. \quad (67)$$

and a sequence projection sums these contributions. Green-image probing uses $g = K^\top h$. All adaptive choices must precede the fresh, prebudgeted block; computed adjoints require outward recurrence-error accounting.

Proof. Conditional on nonzero fixed e , the variables $g_i^\top e/\|e\|$ are iid standard normal. The probability that every absolute projection is below c_δ is $(2\Phi(c_\delta) - 1)^m = \delta$; outward intervals only increase their maximum. Apply the same argument to Ke and use adjointness for the Green-image claim. Since $Kq - \tilde{y} = K(q - q^{[\lambda]}) + Ke_{\text{ar}}$, Eq. (66) follows. The response-free inequality is the triangle bound $\|Kq\|_X \leq \hat{\kappa}(\sigma_{\text{sec}} + \|q^{[\lambda]}\|_U)$. The momentum identity follows by splitting the state probe and writing gradient inner products as directional derivatives. $\square$

Corollary 11 (Corrected-path relinearized Green closure). *Let $v_0 = 0$, $b_j = c_j + v_j$, and*

$$r_j^v = v_{j+1} - J_j v_j - s_j, \quad \bar{s}_j = G(b_j) - b_{j+1}, \quad \bar{J}_j = DG(b_j). \quad (68)$$

Let $\bar{K}_H$ be the anchor-fixed causal Green operator generated by $\bar{J}_0, \dots, \bar{J}_{H-1}$. Assume the pairwise local envelope

$$\|DG(c_j + u) - DG(c_j + w)\| \leq M\|u - w\| \quad (69)$$

whenever $\|u\|, \|w\| \leq \rho$. Suppose $\bar{\kappa} \geq \|\bar{K}_H\|$ and Y is a valid bound, including numerical error, on $\|\bar{K}_H \bar{s}\|_X$. If

$$\bar{D} = 1 - 2\bar{\kappa}MY \geq 0, \quad E = \frac{2Y}{1 + \sqrt{\bar{D}}}, \quad \max_{0 \leq j < H} \|v_j\| + E \leq \rho, \quad (70)$$

then

$$\|x_{a+1:a+H} - b_{1:H}\|_X \leq E. \quad (71)$$

Moreover, the corrected defect obeys the exact cancellation-safe identity

$$\boxed{\bar{s}_j = N_j(v_j) - r_j^v}. \quad (72)$$

Thus an exact variational response ($r^v = 0$) leaves second-order forcing, and the closure $Y + \bar{\kappa}ME^2/2 \leq E$ contains no mixed term. Any verified response, fresh-probe, or Green-image bound from Corollary 10 may supply Y .

Proof. Expanding $G(c_j + v_j)$ around c_j proves Eq. (72). For $e_j = x_{a+j} - b_j$, causal recurrence gives

$$e = \bar{K}_H \bar{s} + \bar{K}_H \bar{N}(e), \quad \bar{N}_j(e_j) = G(b_j + e_j) - G(b_j) - \bar{J}_j e_j. \quad (73)$$

Equation (69) and the domain condition imply $\|\bar{N}(e)\|_U \leq M\|e\|_X^2/2$. Hence $Y + \bar{\kappa}ME^2/2 \leq E$ makes the exact finite-dimensional map self-mapping. Brouwer gives a fixed point, causal forward recurrence makes it unique, and the stable smaller quadratic root is Eq. (70). $\square$

Corollary 12 (Scaled-momentum parameter-forcing closure). *In Corollary 11, let*

$$G(\theta, w) = (\theta - r, r), \quad r = \mu w + \eta \nabla F(\theta), \quad (74)$$

and let $\Pi_\theta(\delta\theta, \delta w) = \delta\theta$. Define the checkpointwise injection

$$\mathcal{B}q_j = (-\eta q_j, \eta q_j) \quad (75)$$

and the anchor shift

$$\mathcal{S}(p_1, \dots, p_H) = (0, p_1, \dots, p_{H-1}). \quad (76)$$

Let Q_0 prepend a zero block to an $(H-1)$ -block forcing sequence and put

$$T_{\theta,0} = \Pi_\theta \bar{K}_H \mathcal{B}Q_0. \quad (77)$$

Suppose, on the declared parameter domains,

$$R_j(u) = \nabla F(\theta_j + u) - \nabla F(\theta_j) - H_j u, \quad \|R_j(u)\| \leq \frac{1}{2} L \|u\|^2. \quad (78)$$

If

$$\kappa_\theta \geq \|T_{\theta,0}\|, \quad Y_\theta \geq \|\Pi_\theta \bar{K}_H \bar{s}\|_X, \quad (79)$$

and

$$D_\theta = 1 - 2\kappa_\theta L Y_\theta \geq 0, \quad E_\theta = \frac{2Y_\theta}{1 + \sqrt{D_\theta}}, \quad (80)$$

with every pointwise parameter ball of radius E_θ inside its declared domain, then

$$\left(\sum_{j=1}^H \|\Pi_\theta(x_{a+j} - b_j)\|^2 \right)^{1/2} \leq E_\theta. \quad (81)$$

For exact operator quantities,

$$\|T_{\theta,0}\|_L \leq \sqrt{2} \eta \|\bar{K}_H\|_L, \quad \|\Pi_\theta \bar{K}_H \bar{s}\|_X \leq \|\bar{K}_H \bar{s}\|_X. \quad (82)$$

Thus the parameter closure is never worse than the full-state corrected closure for a parameter-defined event. Replacing L by checkpointwise L_j and querying $\Pi_\theta \bar{K}_H \mathcal{B} \text{diag}(L_j) Q_0$ gives an equally valid, weakly sharper curvature-profiled coefficient.

Proof. For $e_j = x_{a+j} - b_j$, write $p = (\Pi_\theta e_1, \dots, \Pi_\theta e_H)$. Equation (74) gives the exact remainder factorization

$$G(b_j + e_j) - G(b_j) - \bar{J}_j e_j = \mathcal{B}R_j(\Pi_\theta e_j), \quad (83)$$

with R_j from Eq. (78). Because $e_0 = 0$, the first remainder block is zero. Causal substitution therefore yields the closed parameter equation

$$p = \Pi_\theta \bar{K}_H \bar{s} + T_{\theta,0} R_+(Sp). \quad (84)$$

For $\|p\|_X \leq E$,

$$\|R_+(Sp)\|_U \leq \frac{L}{2} \left(\sum_{j=1}^{H-1} \|p_j\|^4 \right)^{1/2} \leq \frac{L}{2} \|p\|_X^2. \quad (85)$$

Hence the right side of Eq. (84) maps the radius- E_θ parameter ball into itself. Brouwer gives a fixed point; lifting it through $\bar{K}_H$ solves the full causal recurrence, whose forward trajectory from the realized anchor is unique. Equation (80) is the cancellation-safe smaller root. Finally, $\|\Pi_\theta\| = \|Q_0\| = 1$ and $\|\mathcal{B}\| = \sqrt{2}\eta$, proving Eq. (82). The checkpointwise statement follows by absorbing $\text{diag}(L_j)$ into the input side of the same operator. $\square$

Corollary 13 (Deterministic neural-jet release). *In Corollary 11, suppose the optimizer objective is mean cross entropy composed with a neural logit map f_j , plus quadratic weight decay, and the scaled-momentum learning rate is η . On each outer ball $B(c_j, \rho)$ let*

$$\|Df_j\| \leq A_j, \quad \|D^2f_j\| \leq B_j, \quad \|D^3f_j\| \leq C_j. \quad (86)$$

Then Eq. (69) holds with

$$M_{\text{jet}} = \max_{1 \leq j < H} \sqrt{2}\eta \left(2A_j^3 + \frac{3}{2}A_j B_j + \sqrt{2}C_j \right). \quad (87)$$

If the corrected closure issues with this M_{jet} and $p = \max_j \|v_j\|$, every true-class-versus-competitor margin satisfies

$$|m_{iq}(x_{a+j}) - m_{iq}(c_j)| \leq \sqrt{2}A_j(p + E). \quad (88)$$

Strict event logic based on Eq. (88) therefore needs no randomized output-Jacobian operator. A policy may try this deterministic release first and instantiate the sharper output probe only after failure; such predictable fallback adds no probability charge to cases released by the deterministic screen and preserves the preallocated family bound otherwise.

Proof. For cross entropy, $\|D\ell\| \leq \sqrt{2}$, $\|D^2\ell\| \leq 1/2$, and $\|D^3\ell\| \leq 2$. The third-order chain rule gives

$$\|D^3(\ell \circ f_j)\| \leq 2A_j^3 + 3(1/2)A_jB_j + \sqrt{2}C_j. \quad (89)$$

Averaging examples cannot increase the norm, and quadratic weight decay has zero third derivative. The scaled-momentum Jacobian changes by at most $\sqrt{2}\eta$ times the objective-Hessian change, proving Eq. (87). The corrected tube lies in the declared outer ball because $p + E \leq \rho$. The mean-value theorem bounds each logit change by $A_j(p + E)$; subtracting two logits gives Eq. (88). Proposition 2 supplies the fallback accounting. $\square$

Corollary 14 (Nested-prefix direct-image/Gram release). *Let T be fixed before iid $g_i \sim \mathcal{N}(0, I)$ are drawn, and fix $1 \leq m_1 < \dots < m_L$, integers $q_\ell \geq 1$, and $\delta_\ell > 0$ with $\sum_\ell \delta_\ell \leq \delta$. Put*

$$c_\ell = \Phi^{-1}\left(\frac{1 + \delta_\ell^{1/m_\ell}}{2}\right). \quad (90)$$

Then with probability at least $1 - \delta$, simultaneously for every prefix,

$$\|T\| \leq \frac{\max_{i \leq m_\ell} \|Tg_i\|}{c_\ell}, \quad \|T\| \leq \left(\frac{\max_{i \leq m_\ell} \|(T^T T)^{q_\ell} g_i\|}{c_\ell}\right)^{1/(2q_\ell)}. \quad (91)$$

Hence one may inspect the direct release, apply T^T to the cached images only if needed, and stop at the first issuing prefix. For a predeclared operator family, summing $\delta_{n,\ell}$ over operators and prefixes gives the family-wise bound.

Proof. For a unit top right singular vector v and $Tv = \|T\|u$, $\|Tg_i\| \geq \|T\|\|v^T g_i\|$ and $\|(T^T T)^{q_\ell} g_i\| \geq \|T\|^{2q_\ell} \|v^T g_i\|$. Prefix ℓ fails only if all m_ℓ absolute standard-normal projections are below c_ℓ , an event of probability δ_ℓ . A union bound proves simultaneous validity; direct screening, Gram fallback, and stopping are functions of already valid bounds. $\square$

Proposition 5 (Causal prefix commutation). *Let $A_H(x_0)$ be the horizon- H affine reference and R_H one anchor-fixed variational recentering sweep. If P_h retains states $0:h$ and neither construction truncates before $h \leq H$, then*

$$P_h A_H(x_0) = A_h(x_0), \quad P_h R_H(c) = R_h(P_h c), \quad P_h R_H^S A_H(x_0) = R_h^S A_h(x_0). \quad (92)$$

All S sweeps may therefore be pipelined in one forward pass with $O(Hd + Sd)$ rather than $O(SHd)$ path storage, and construction may stop when a sealed horizon or first persistent predicted event is reached.

Proof. The affine state at $j + 1$ uses only its state at j . In one recentering sweep, $d_{j+1} = DG(c_j)d_j + G(c_j) - c_{j+1}$ depends only on (c_j, c_{j+1}, d_j) . Induction in time proves both prefix identities; induction over the sweep index proves the last. Pipelining retains the preceding sweep's current/next states and each sweep's current correction, while storing only the final path. $\square$

Proposition 6 (Persistent first-passage bracket). *If $n_j^- \leq n_j \leq n_j^+$ for $0 \leq j \leq H$ and the finite endpoints $L, U \in \{0, \dots, H - K + 1\}$ are defined by Eq. (27), then $T_{r,K}$ is finite and $L \leq T_{r,K} \leq U$.*

Proof. Before L , every length- K block contains an offset with $n^+ < r$, so the true block cannot be entirely above gate. At U , every offset has $n^- \geq r$, so the true block is above gate throughout. $\square$

Proposition 7 (Witness-sparse, predictable, role-fused event transport). *Let P_j certify that the gate holds at j , and F_j certify that it fails. A bracket $[L, U]$ is valid if P_j holds for $j = U, \dots, U + K - 1$ and a set W^- of certified failure times intersects every interval $[t, t + K - 1]$ for $t < L$. For a fixed set of available failure times, greedily selecting the rightmost available point in the earliest-ending uncovered interval gives a minimum-cardinality W^- . If the optimizer depends on training outputs f_{tr} and the event on evaluation outputs f_{ev} , it is sufficient to bound derivatives of f_{tr} at transition-input states and derivatives of f_{ev} only at the success and failure-witness times. Witness times may be acquired predictably from centerline margins and prior certificate results using fresh prebudgeted probe blocks. At a time requiring both roles, one bound on $D(f_{\text{tr}} \oplus f_{\text{ev}})$ may replace the two component queries.*

Proof. The success window proves $T_{r,K} \leq U$, while a certified failure in every earlier length- K interval proves $T_{r,K} \geq L$. For interval stabbing, replace the first point of any optimal solution that hits the earliest-ending uncovered interval by that interval’s rightmost available point; this cannot uncover a later-ending interval. Induction proves greedy optimality. Finally, optimizer drift factors only through the objective and f_{tr} , whereas event margins factor only through f_{ev} at the logical witnesses. Predictable acquisition is covered by Proposition 2. Finally, each component Jacobian is an output projection of the direct-sum Jacobian, so its norm is no larger. Intersecting the prebudgeted operator events completes the claim. $\square$

Affine launch-cost consequence. Let $Q \subseteq \{1, \dots, H\}$ be the q event-query times, and let one call on n examples cost $a + bn$. Dense, separate-role, and fused costs are respectively

$$\begin{aligned} C_d &= Ha + bHn_{\text{all}}, \\ C_s &= (H - 1 + q)a + b\{(H - 1)n_{\text{tr}} + qn_{\text{ev}}\}, \\ C_f &= (H - 1 + \mathbf{1}_{\{H \in Q\}})a + b\{(H - 1)n_{\text{tr}} + qn_{\text{ev}}\}. \end{aligned}$$

Thus fusion removes exactly $(q - \mathbf{1}_{\{H \in Q\}})a$ from the separate path. Separate roles beat the dense baseline only if $b[Hn_{\text{all}} - (H - 1)n_{\text{tr}} - qn_{\text{ev}}] > (q - 1)a$. This explains the observed launch-overhead regression without asserting a hardware-independent runtime guarantee.

Lemma 1 (Scaled-momentum derivative drift). *If $\|\nabla^2 F(\theta + u) - \nabla^2 F(\theta)\| \leq L \|u\|_2$, then the map in Eq. (22) satisfies $\|DG(x + h) - DG(x)\| \leq \sqrt{2}\eta L \|h\|_2$.*

Proof. Only the parameter-to-update Hessian block changes. Applied to $(d\theta, dw)$, the Jacobian difference is $(-\eta\Delta H d\theta, \eta\Delta H d\theta)$, whose norm is at most $\sqrt{2}\eta L \|h\|_2 \|(d\theta, dw)\|_2$. $\square$

B Complete analytic Transformer derivative envelope

This section derives the deterministic first-, second-, and third-order neural envelopes used in every Transformer certificate. Random-direction tests are useful regression checks, but they are not part of the validity argument. The argument below follows the computational graph line by line.

B.1 Norms, parameter blocks, and diagonal jets

Every activation tensor is flattened and given its Euclidean (equivalently, Frobenius) norm. Matrix parameters use the Frobenius norm as coordinates and the spectral norm when acting on an activation. The complete parameter vector is partitioned into 14 orthogonal blocks:

b	block	contents
0	embedding	token and position embeddings jointly
1–3	attention weights	W_Q, W_K, W_V (disjoint row slices)
4–6	attention biases	b_Q, b_K, b_V (disjoint slices)
7–8	attention output	W_O, b_O
9–10	feed-forward 1	W_1, b_1
11–12	feed-forward 2	W_2, b_2
13	readout	W_R

The grouped/sliced blocks are still orthogonal: the squared block Frobenius norms sum to the squared norm of the original flat perturbation. Hence a unit direction u has block radii $s_b = \|u_b\|_2$ satisfying $\sum_b s_b^2 = 1$.

For a smooth tensor-valued stage f , write $P_k^f(s)$ for a polynomial with nonnegative coefficients such that

$$\|D^k f(\theta)[u, \dots, u]\|_2 \leq P_k^f(s), \quad k = 1, 2, 3. \quad (93)$$

The restriction to a repeated direction loses nothing for the Euclidean multilinear norm. Indeed, each scalarization of $D^k f$ is a symmetric k -linear form on a real Hilbert space; Banach's theorem gives equality between its multilinear norm and the norm of its diagonal homogeneous polynomial. Dualizing the output gives the same result for tensor-valued f [Carando and Rodríguez, 2019]. Thus a valid bound on Eq. (93), maximized over $\sum_b s_b^2 \leq 1$, bounds the full Fréchet derivative norm.

For a monomial $m(s) = \prod_b s_b^{\alpha_b}$ of degree $d = \sum_b \alpha_b$, Lagrange multipliers give the exact sphere maximum

$$\sup_{\sum_b s_b^2 \leq 1} m(s) = \prod_{b: \alpha_b > 0} \left(\frac{\alpha_b}{d}\right)^{\alpha_b/2}. \quad (94)$$

We maximize a nonnegative polynomial by summing these exact monomial maxima. This can be conservative for a sum, but it is always valid and never exceeds the older scalar rule that sets every $s_b = 1$.

B.2 Primitive jet rules

Represent a stage by $\mathcal{J}(f) = (V_f, P_1^f, P_2^f, P_3^f)$, where $V_f \geq \|f\|_2$ on the relevant input set. All inequalities below follow from the triangle inequality, submultiplicativity of Frobenius/spectral norms, and the ordinary Fréchet product and chain rules.

For addition, $h = f + g$,

$$V_h = V_f + V_g, \quad P_k^h = P_k^f + P_k^g. \quad (95)$$

For a bilinear contraction $h = \alpha f g$ (matrix multiplication included),

$$\begin{aligned} V_h &= |\alpha| V_f V_g, \\ P_1^h &= |\alpha| (P_1^f V_g + V_f P_1^g), \\ P_2^h &= |\alpha| (P_2^f V_g + 2P_1^f P_1^g + V_f P_2^g), \\ P_3^h &= |\alpha| (P_3^f V_g + 3P_2^f P_1^g + 3P_1^f P_2^g + V_f P_3^g). \end{aligned} \quad (96)$$

For an elementwise or rowwise smooth map $h = \phi \circ f$ with $\|D^k \phi\|_2 \leq a_k$ on the stage range,

$$\begin{aligned} P_1^h &= a_1 P_1^f, \\ P_2^h &= a_1 P_2^f + a_2 (P_1^f)^2, \\ P_3^h &= a_1 P_3^f + 3a_2 P_1^f P_2^f + a_3 (P_1^f)^3. \end{aligned} \quad (97)$$

Products of the P_k in Eqs. (96)–(97) are polynomial convolutions over block labels, so mixed derivatives and their combinatorial multiplicities are retained exactly before the sphere relaxation.

For a parameterized affine stage with sequence length ℓ ,

$$h(X, W, b) = XW^\top + \mathbf{1}_\ell b^\top, \quad (98)$$

we assign the weight jet $(\|W\|_{\text{op}}, s_W, 0, 0)$ and bias jet $(\sqrt{\ell} \|b\|_2, \sqrt{\ell} s_b, 0, 0)$ and apply Eq. (96) plus Eq. (95). This is valid because

$$\|XW^\top\|_F \leq \|X\|_F \|W\|_{\text{op}}, \quad \|\mathbf{1}_\ell b^\top\|_F = \sqrt{\ell} \|b\|_2. \quad (99)$$

On a radius- R parameter ball we replace $\|W\|_{\text{op}}$ by $\|W_0\|_{\text{op}} + R$ and $\|b\|_2$ by $\|b_0\|_2 + R$; using the full radius for each value bound is conservative even though the derivative polynomial retains the sharper block allocation.

B.3 Global activation constants

The Gaussian-error linear unit is $\text{GELU}(x) = x\Phi(x)$, with standard-normal density φ . Direct differentiation gives

$$\begin{aligned} g'(x) &= \Phi(x) + x\varphi(x), \\ g''(x) &= (2 - x^2)\varphi(x), \\ g'''(x) &= (x^3 - 4x)\varphi(x). \end{aligned} \tag{100}$$

Elementary one-dimensional maximization yields

$$\sup |g'| < 1.13, \quad \sup |g''| < 0.80, \quad \sup |g'''| < 2. \tag{101}$$

For an elementwise map these scalar constants also bound the induced Euclidean multilinear derivatives, since $\|x_1 \odot \cdots \odot x_k\|_2 \leq \prod_i \|x_i\|_2$.

For $p = \text{softmax}(x)$, let $C(p) = \text{diag}(p) - pp^\top$. The categorical covariance bound gives $\|C(p)\|_{\text{op}} \leq 1/2$. If $q = C(p)h$, then

$$DC(p)[h] = \text{diag}(q) - qp^\top - pq^\top, \tag{102}$$

whose norm is at most $3\|h\|_2/2 < 2\|h\|_2$. Differentiating once more and using $\|p\|_2 \leq 1$, $\|q\|_2 \leq \|h\|_2/2$, and $\|DC(p)[k]\|_2 \leq 3\|k\|_2/2$ gives the conservative bound $5 < 6$ for the third derivative. Therefore the implementation uses

$$(a_1, a_2, a_3)_{\text{softmax}} = (1/2, 2, 6). \tag{103}$$

The causal mask is a constant translation before softmax and contributes no derivative. Rowwise application is block diagonal, so the same constants hold under the flattened Frobenius norm.

B.4 One-block causal Transformer graph

The certified architecture has sequence length three, one normalization-free causal attention block, one GELU feed-forward block, residual additions, and a linear readout. For each of the finite p^2 input pairs the implementation applies the following graph.

Embedding. Token lookup and positional addition are linear. A sequence may repeat an input token, and all three positions vary. The joint lookup operator is bounded by $\sqrt{3}$, so

$$\mathcal{J}(X_0) = (V_{X_0}, \sqrt{3}, s_{\text{emb}}, 0, 0). \tag{104}$$

Queries, keys, and values. For $A \in \{Q, K, V\}$,

$$A = X_0 W_A^\top + \mathbf{1}_3 b_A^\top. \tag{105}$$

The three weight slices and three bias slices are disjoint blocks. Each jet is given by Eqs. (96) and (99).

Scores and attention. Reshaping into heads is an isometry. With head dimension d_h ,

$$S = QK^\top / \sqrt{d_h}, \quad P = \text{softmax}_{\text{row}}(S + \mathcal{M}), \quad A = PV. \tag{106}$$

The score and attended-value jets use Eq. (96); the probability jet uses Eqs. (97) and (103). Matrix contraction is safe because $\|QK^\top\|_F \leq \|Q\|_F \|K\|_F$ and $\|PV\|_F \leq \|P\|_F \|V\|_F$.

Output projection, residuals, feed-forward, and readout. The remaining graph is

$$\begin{aligned} X_1 &= X_0 + (AW_O^\top + \mathbf{1}_3 b_O^\top), \\ U &= X_1 W_1^\top + \mathbf{1}_3 b_1^\top, \quad V = g(U), \\ X_2 &= X_1 + (VW_2^\top + \mathbf{1}_3 b_2^\top), \\ f_\theta(x) &= (X_2)_{3,:} W_R^\top. \end{aligned} \tag{107}$$

Affine stages use Eq. (98), residuals use Eq. (95), and GELU uses Eqs. (97) and (101). Selecting the last token is nonexpansive. This completes the line-by-line derivation of the three output derivative polynomials. For hash-preserving reproducibility, the sealed implementation upper-bounds the bias-free readout by the larger affine family that also permits an auxiliary bias perturbation. This is a model-class relaxation, not a stated graph component: algebraically it adds one nonnegative degree-one monomial and changes no higher derivative polynomial. Deleting that monomial gives the exact bias-free graph and weakly decreases every downstream bound, so all sealed certificates remain valid. A released regression test checks this identity in both scalar and block-polynomial implementations.

B.5 Making center values valid on the complete ball

The input set is finite, so every center-stage maximum V_i^0 is evaluated exactly over all p^2 pairs in the real-arithmetic specification. Center values alone are not ball-valid. Let $B_{1,i}(V^0 + \Delta, R)$ be the first derivative polynomial for stage i after substituting proposed inflated stage values. All its coefficients are nonnegative and it is coordinatewise nondecreasing. We seek a post-fixed inflation

$$\Delta_i \geq RB_{1,i}(V^0 + \Delta, R) \quad \text{for every stage } i. \quad (108)$$

Lemma 2 (Ball-valid value chain). *If Eq. (108) holds, then every stage value on $B(\theta_0, R)$ is bounded by $V_i^0 + \Delta_i$, and the derivative polynomials computed with those values enclose $D^k f$ for $k = 1, 2, 3$ throughout the ball.*

Proof. Proceed in computational-graph order. For any $\theta_0 + h$ with $\|h\|_2 \leq R$, the integral mean-value identity and the inductively valid upstream stage values give

$$\|f_i(\theta_0 + h)\|_2 \leq V_i^0 + RB_{1,i}(V^0 + \Delta, R) \leq V_i^0 + \Delta_i. \quad (109)$$

The primitive rules above then apply with valid values at every stage. Their composition encloses all derivatives through order three. $\square$

The implementation starts at $\Delta = 0$ and repeatedly replaces each coordinate by the maximum of its current value and the right side of Eq. (108). This is monotone iteration. The frozen Transformer records use a relative 10^{-9} binary64 consistency tolerance, so this check is part of their disclosed numerical rather than exact-real layer. A post-seal strict wrapper instead inflates each iterate with directed binary64 `nextafter` steps until every displayed inequality is literally satisfied. On the shortest sealed issuance, recomputation finds 338 frozen stage inequalities with positive but tiny deficits (maximum 1.90×10^{-23} absolute and 5.15×10^{-14} relative); the strict wrapper removes all of them, leaves the rounded derivative jets unchanged, and reproduces the issued bracket. This hardens the internal post-fixed check but is not a substitute for outward real arithmetic. The same positivity gives a useful monotonicity fact.

Lemma 3 (Radius monotonicity). *For $0 \leq R_1 \leq R_2$, the analytic post-fixed envelopes may be chosen so that $B_k(R_1) \leq B_k(R_2)$ for $k = 1, 2, 3$.*

Proof. Every primitive coefficient, value inflation, and weight/bias enlargement is nonnegative and nondecreasing in R . A post-fixed point for R_2 is therefore also an upper solution for the R_1 inequalities. Substitution into the nonnegative derivative polynomials proves the claim. $\square$

Lemma 3 is what makes the strong unsigned baseline audit decisive: once its required radius contains the signed ball, failure even with the smaller-ball M rules out closure under the same analytic envelope.

B.6 Cross-entropy and optimizer-map drift

For one example, cross-entropy is $\ell_y(z) = -z_y + \log \sum_q e^{z_q}$. With $p = \text{softmax}(z)$,

$$\|D\ell_y\|_2 = \|p - e_y\|_2 \leq \sqrt{2}, \text{quad} \|D^2\ell_y\|_2 = \|C(p)\|_2 \leq \frac{1}{2}. \quad (110)$$

The same calculation used above gives the generous global bound $\|D^3\ell_y\|_2 \leq 2$. If B_1, B_2, B_3 enclose the first three derivatives of the per-example logit map on the parameter ball, the third-order chain rule therefore gives

$$\|D^3(\ell_y \circ f_\theta)\|_2 \leq 2B_1^3 + \frac{3}{2}B_1B_2 + \sqrt{2}B_3. \quad (111)$$

An average over examples obeys the same maximum bound. Weight decay is quadratic and contributes no Hessian drift. The center logit Jacobian is bounded separately by the matrix-free Gram probe; if its bound is $\hat{B}_{1,0}$, then

$$B_1(R) = \hat{B}_{1,0} + B_2(R)R \quad (112)$$

is ball-valid. Substitution into Eq. (111) yields the objective Hessian-Lipschitz constant L .

In scaled momentum coordinates $x = (\theta, w = \eta v)$,

$$G(\theta, w) = (\theta - r, r), \quad r = \mu w + \eta \nabla F(\theta). \quad (113)$$

Only the Hessian block changes with the state. Consequently

$$\|DG(x+h) - DG(x)\|_2 \leq \sqrt{2}\eta L \|h\|_2, \quad (114)$$

which is the implemented $M = \sqrt{2}\eta L$. The anchor error is exactly zero, so only transitions leaving $c_1, \dots, c_{H-1}$ enter the maximum.

B.7 Output-margin transport

The parameter component of a scaled-state perturbation has norm at most the full state perturbation. Taylor’s theorem gives, for each logit vector,

$$\|f(\theta+h) - f(\theta)\|_2 \leq B_1R + \frac{1}{2}B_2R^2. \quad (115)$$

A true-versus-competitor margin is the inner product with $e_y - e_q$, whose norm is $\sqrt{2}$. Thus every margin changes by at most

$$E = \sqrt{2} \left(B_1R + \frac{1}{2}B_2R^2 \right). \quad (116)$$

Strictly positive lower margins certify correct examples. A strictly negative upper margin certifies an incorrect example. Counting these statements gives the lower/upper accuracy paths used by the persistent first-passage corollary.

C Randomness and finite-precision claim boundary

C.1 What the Gaussian theorem randomizes

Proposition 1 is an exact-arithmetic statement about independent $N(0, I)$ vectors. Its probability is over probe randomness only—not over training seeds, examples, candidate selection, or future outcomes. The frozen protocol declares the complete operator family before training, divides the 10^{-6} family budget over that maximum family, and makes no randomized query during screening.

The executable uses an OS-random master nonce frozen before training. Each complete operator identity is domain-separated by SHA-256 into a 63-bit seed; a registry rejects unknown identities, repeated queries, and seed collisions. PyTorch then maps that seed deterministically to pseudorandom normal variates. Domain separation prevents accidental stream reuse, but deterministic PRNG outputs are not literally independent mathematical Gaussians. Accordingly, the Transformer statement is a family-wise high-confidence numerical claim under the standard ideal-PRNG model. It is not an unconditional theorem about the fixed bit strings produced by a particular generator. A deterministic replacement would require an independently certified spectral-norm enclosure.

C.2 Robust theorem for computational error

The abstract Green theorem already admits explicit computational remainder. If $\tilde{s}$ approximates the defect and a computed response $\tilde{z}_0 = 0$ has exact-real recurrence residual

$$d_j = \tilde{z}_{j+1} - J_j \tilde{z}_j - \tilde{s}_j, \quad (117)$$

then

$$\|s - \tilde{s}\|_U \leq \sigma, \quad \|d\|_U \leq \tau \implies \|K_H s - \tilde{z}\|_X \leq \kappa(\sigma + \tau). \quad (118)$$

Using $\alpha = \kappa(\sigma + \tau)$ in Theorem 1 is sufficient. Likewise, outward upper bounds on κ, M, B_1, B_2 and outward lower bounds on strict margin slacks preserve the event certificate.

The Transformer implementation evaluates the neural jets, HVP/VJP products, Gram iterates, and margins in deterministic float64 and does not yet supply nonzero verified σ, τ for every operation. Its 9/9 and 11/11 results therefore remain high-confidence numerical certificates rather than computer-assisted exact-real proofs. The fixed-radius cohort has minimum closure slack 0.1445 and minimum strict output slack 2.67×10^{-5} . In contrast, the WDBC and digits tanh confirmations enclose all 63 issued events with direct 192-bit outward continuations whose recurrences are independent of the randomized Green event; the dense modular study provides an additional outward-verified regime. These direct passes validate the retained exact-real event brackets, not every Green inequality. This distinction is stated in the abstract, results table, and limitations.

D Complete frozen protocols

WDBC. The Wisconsin Diagnostic Breast Cancer data contain 569 labeled examples with 30 real-valued features [Wolberg et al., 1993, Street et al., 1993]. Fresh seeds 101–124 use class-stratified 341/113/115 train/trigger/certification splits and train-only standardization. A width-8 tanh MLP (266 parameters) runs 1,000 full-batch float64 GD updates with $\eta = 0.005$ and weight decay 0.001. At 90/92.5/95% gates with $K = 10$, the selector freezes the first five-update checkpoint with train accuracy at least 80% and trigger accuracy in $[\tau - 10\%, \tau)$. Its API cannot materialize certification rows; candidates and certificates are sealed before future trajectories are joined. The method fixes $H = 300$, four sweeps, $m = 8, q = 4$, a 10^{-6} family budget, and the smallest closure root. Inspected development seeds 0–7 fixed every choice before the method seal. The separate 192-bit verifier begins at zero radius on each exact dyadic anchor and propagates a direct one-step tube without a Green norm or randomized probe.

Digits. The scikit-learn digits data contain 1,797 normalized 8×8 images from the UCI Optical Recognition test set [Alpaydin and Kaynak, 1998]. The protocol freezes digit parity, seeds 501–512, parity-stratified 1,077/359/361 splits, train-only standardization, and a width-8 tanh MLP (538 parameters) trained by 600 full-batch GD updates with $\eta = 0.03$. The 90/92.5% gates use $K = 10$, the WDBC anchor rule, $H = 400$, three sweeps, $m = 8, q = 4$, and a 10^{-6} family budget. Inspected development seeds 0–2 fix the sweep budget; signed and unsigned methods receive the same centerline. Code, fresh seeds, and the primary signed-only endpoint are sealed before training, under the WDBC selection/outcome barrier. The post-outcome 192-bit verifier directly encloses the exact-real optimizer map without a Green norm or probe.

Fixed-radius Transformer cohort. Fresh seeds 331–354 train one-block causal Transformers on addition modulo 17: model dimension 32, four heads, GELU width 128, residuals, no normalization or dropout, and 13,792 parameters. The 27,584-dimensional momentum state runs 6,000 full-batch float64 updates with $\eta = 0.01, \mu = 0.9$, and weight decay 0.01; each seed splits 289 examples into 173/58/58. The protocol fixes 70/80/90% gates, $K = 25, H = 300$, and four sweeps. A 40-update causal scan freezes the first predicted event after train accuracy reaches 99%, trigger accuracy reaches $\max(50\%, \tau - 20\%)$, and the current certification count lies one to three examples below gate. It fixes $R = 2Z, m = 16, q = 8$, and a 10^{-6} budget over 21,744 possible operators. Future trajectories remain inaccessible through certificate sealing. This selector reads the current certification count, although the certification set is disjoint from trigger; hence the cohort is prospective and outcome-sealed, not untouched by selection.

Response-centered Transformer cohort. Fresh seeds 355–378 repeat the model, optimizer, splits, gates, and outcome barrier. Before these runs, code and constants are sealed for four variational sweeps, outer domain $\rho = 2Z$, the response-centered root in Corollary 1, and a shared-probe ladder $q = 1, \dots, 8$. The first power whose nonlinear closure and persistent-event margins both pass issues. A response-aware lower bound $\kappa \geq \max(1, Z/\|s\|_X)$ permits zero-query abstention; the matched baseline reuses the power-8 quantities with radius $2Z$. The family budget is 10^{-6} over the worst-case 24-by-3 candidate universe. One selector/constructor horizon mismatch among 19 sealed candidates

is retained as an outcome-blind construction abstention under the pre-outcome execution amendment in Appendix H.

Diagnostic and numerical studies. Three earlier Transformer candidates (seeds 321–322) with already known outcomes supply the excluded 0–4-sweep audit. The post-seal unsigned audit replaces $Z = \|K_H s\|_2$ by $Z_u = \kappa \|s\|_2$ while preserving the right inverse, outer-ball envelope, and smallest closure root; it is a matched scalar Euclidean comparator, not a claim about componentwise or Taylor-model validators. Other post-seal diagnostics batch the same 16 committed probes and test exact products for a two-block LayerNorm/AdamW map. Dense mod-11 MLPs (827 parameters) provide 192-bit outward evaluation of every issued inequality, and a disjoint mod-13 study (1,933 parameters) supplies HVP-only active/complement uncertainty. Direct 192-bit continuations independently retain every issued WDBC and digits bracket.

E Complete WDBC confirmation and outward audit

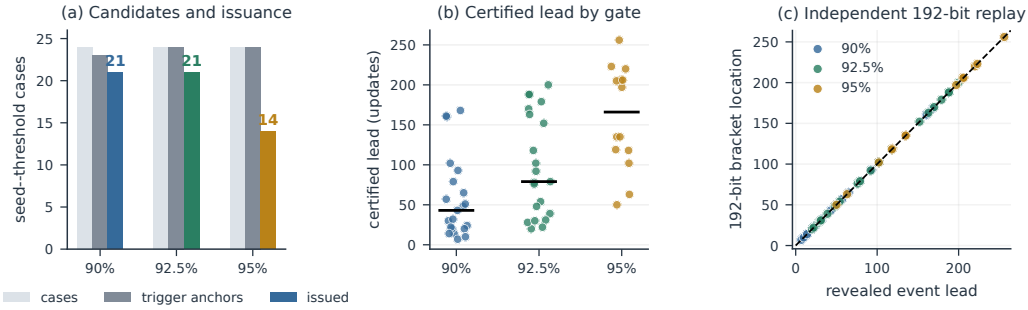


Figure 3: Fully frozen WDBC confirmation. (a) Candidate and issuance counts remain high as the gate rises. (b) Higher transitions are certified farther in advance. (c) All 56 outward-retained singleton brackets equal the subsequently revealed crossing; jitter only separates coincident points.

The downloaded WDBC CSV has SHA-256 prefix `FF5FC1F247213DB`. The frozen method, candidate manifest, certificate manifest, and final independent audit have SHA-256 prefixes `3408AF37C0395926`, `F552930042296C89`, `0444FE25E4C414B9`, and `54DA92445CC7E909`, respectively; full hashes are in the artifact manifest. The source seal covers 16 protocol and implementation files. The independent auditor verifies the complete hash chain, reconstructs every checkpoint exactly, recomputes candidate and issuance dispositions, and evaluates every persistent bracket without importing the experiment’s aggregation helper.

The 72-case candidate disposition is 71 frozen trigger-only anchors and one case with no anchor. The certificate disposition is 56 issued and 15 abstentions because the frozen centerline has no future event on the newly opened certification split. The 56 issued cases query 56 Green operators; the original Green route has realized ideal-Gaussian union bound 7.78×10^{-7} . Minimum float64 closure and output slacks are 0.9999999907 and 2.3753122×10^{-5} , respectively. The realized sequence error differs from the radius by at most 1.27×10^{-14} in absolute terms; the outward pass resolves this binary64 diagnostic rather than treating a strict floating-point comparison as proof.

The 192-bit verifier is a separately hashed direct validated-continuation program. It treats every stored binary64 datum as an exact dyadic input and outward-encloses tanh network evaluation, exact-Hessian construction, eigenpair residuals, optimizer Jacobians, one-step defect recurrences, nonlinear drift, and event margins. Its anchor radius is zero and its recurrence contains no Green norm, probe, or PRNG call; therefore the Gaussian event above is not a condition of an outward-retained bracket. Across 40 unique seed-anchor tubes it verifies 5,089 state transitions. The largest optimizer-Jacobian norm is 1.00744, the largest Hessian-evaluation enclosure error is 3.74×10^{-12} , and the largest eigen-numerical error is 5.87×10^{-11} . All 56 singleton brackets remain identical and all 56 cover.

Table 3: WDBC confirmation by prespecified gate. Coverage is conditional on issuance; each issued bracket is a singleton.

Gate	Candidates	Issued	Covered	Median lead	Max lead
90%	23/24	21	21/21	43	168
92.5%	24/24	21	21/21	79	200
95%	24/24	14	14/14	166	256
All	71/72	56	56/56	79	256

Table 4: Post-seal matched WDBC validated-numerics comparison on the 56 Green-issued coordinates. Direct timing excludes construction of the shared four-sweep centerline, biasing the runtime ratio in its favor.

Validator	Events	Covered	Random condition	Aggregate time
GREENCERT event records	56	56/56	Gaussian Green bound	33.88 min
Direct 192-bit continuation	56	56/56	none	7.12 h

The direct/GreenCert aggregate ratio is $12.61\times$. The direct comparator was not run on the 15 Green abstentions, so Table 4 compares matched rigor and cost rather than unconditional availability. Both methods use the same four-sweep reference; only the direct one-step tube dispenses with the Green closure. Exact-real scope begins at the stored checkpoint and does not retroactively verify the PyTorch/BLAS training execution.

F Complete prospective digits audit

The bundled scikit-learn digits arrays have SHA-256 F6D9E39F37DC45D3. The method, candidate, and pre-outcome certificate seals have prefixes 9901CD1A0E3FEABF, BFA31D77050F164D, and 99AC9C628AFB4FC9. The method seal predates every seed 501–512 artifact; the candidate seal predates the first full-split loader call; and the certificate seal records seven signed, six unsigned, and one signed-only issuance before any future certification trajectory exists. Reconstruction after the seal has zero checkpoint error.

All 24 seed–gate cases become candidates. Twelve reach a Green query and consume a realized ideal-Gaussian family budget of 5×10^{-7} . Seven signed certificates issue across six seeds and all seven cover. Table 5 records the complete issued set; the supplement records all 17 abstentions. The primary signed-only event has ample separation: $0.2369 < 1$ signed versus $13.018 > 1$ unsigned, with strict output slack 1.43×10^{-5} .

A post-seal verifier independently rebuilds each three-sweep path and propagates a zero-radius 192-bit Arb tube under the exact-real full-batch map. It imports neither the Green radius nor its Gaussian probe. All seven distinct seed–anchor tubes reach their horizons; all seven outward brackets are identical to Table 5 and cover. Maximum radius is 5.07×10^{-9} , minimum strict logic slack is 1.46×10^{-5} , maximum Hessian interval-row radius is 2.18×10^{-11} , and maximum eigensolver numerical error is 1.39×10^{-10} . This includes the signed-only [147, 147] event.

Table 5: All issued certificates in the prospective non-modular digits study. Lead is relative to the frozen anchor; every bracket contains the revealed event. U marks whether the matched unsigned method also issues.

Seed	Gate	Anchor	Lead	Bracket	Signed stat.	Unsigned stat.	U
503	90%	35	49	[49,49]	4.9×10^{-6}	9.0×10^{-5}	yes
504	90%	60	49	[49,49]	1.9×10^{-8}	5.0×10^{-7}	yes
506	90%	20	70	[70,70]	2.1×10^{-5}	4.4×10^{-4}	yes
507	90%	35	117	[117,117]	0.0137	0.2944	yes
509	90%	35	147	[147,147]	0.2369	13.018	no
512	90%	20	42	[42,42]	3.5×10^{-8}	7.8×10^{-7}	yes
512	92.5%	35	73	[73,73]	2.8×10^{-4}	0.0086	yes

The post-seal selectivity audit also compares every finite frozen prediction with its revealed event. Table 6 gives the only two inaccurate finite centerlines. Both were rejected by deterministic early closure before an outcome or randomized probe was available.

Table 6: Every inaccurate finite forecast in the prospective digits study. Error is prediction minus event; the lower bound is the favorable $\kappa \geq 1$ closure statistic, so a value above one proves abstention without querying the Green norm.

Seed	Gate	Anchor	Prediction	Event	Error	Lower bound	Disposition
510	92.5%	45	218	219	-1	6,512	early abstain
511	90%	50	358	361	-3	4,505	early abstain

G Complete fresh Transformer audit

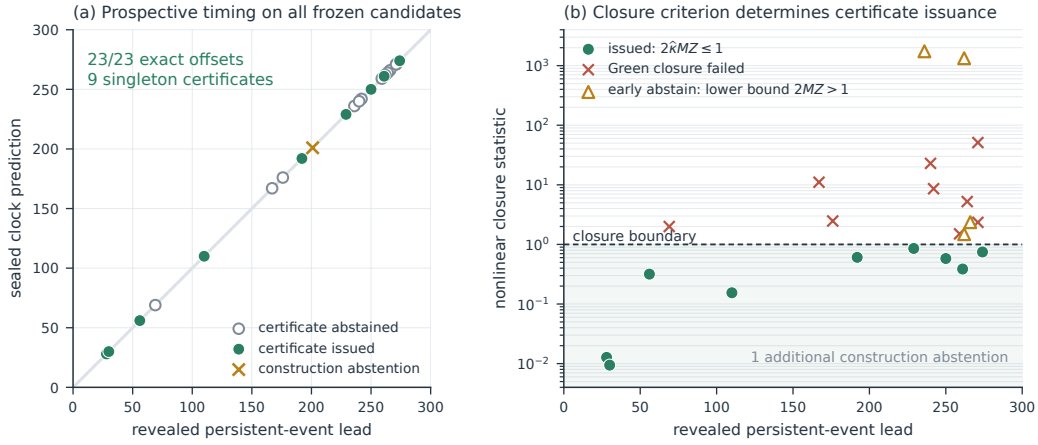


Figure 4: Prospective fixed-radius Transformer confirmation. (a) Every one of the 23 prospectively selected clock events matches the revealed offset; filled points are the nine issued certificates. (b) The nonlinear closure criterion determines certificate issuance. Triangles are proof-preserving early lower bounds; one construction abstention has no statistic.

Table 7 reports every frozen candidate, including abstentions. For Green-probed rows, “stat.” is $2\kappa MZ$; for early rows it is the lower bound $2MZ$. The construction row has no valid centerline and no statistic. Prediction and revealed lead agree in all rows.

Table 7: Complete frozen fresh Transformer candidate audit. “Stat.” is the full Green closure for probed rows and its $\kappa \geq 1$ lower bound for early abstentions.

Seed	Gate	Anchor	Lead	Disposition	Stat.
331	70%	1,160	262	early abstain	1.485
331	80%	1,520	271	closure abstain	51.219
333	70%	3,000	28	issued	0.013
333	80%	3,080	192	issued	0.607
335	70%	2,440	201	construction abstain	–
335	80%	2,880	110	issued	0.154
335	90%	3,240	229	issued	0.855
337	70%	1,080	242	closure abstain	8.603
337	80%	1,720	250	issued	0.580
342	70%	1,480	69	closure abstain	1.996
345	70%	960	236	early abstain	1,739.181
345	80%	1,320	30	issued	0.009
345	90%	2,120	261	issued	0.386
347	70%	2,000	266	early abstain	2.362
348	70%	2,200	56	issued	0.318
348	80%	2,840	264	closure abstain	5.240
348	90%	3,240	176	closure abstain	2.471
350	70%	1,440	262	early abstain	1,331.466
350	80%	2,920	274	issued	0.747
351	70%	1,600	167	closure abstain	11.062
351	80%	2,200	240	closure abstain	22.905
352	80%	5,320	259	closure abstain	1.505
354	70%	2,120	271	closure abstain	2.346

The 72-case disposition is: 23 candidates, 29 causal screens with no future modal candidate, 19 never eligible, and one gate already present. Candidate rate is 31.9%; issuance is 12.5% over all cases and 39.1% conditional on a candidate. Candidate seeds are {331,333,335,337,342,345,347,348,350,351,352,354}; issuing seeds are {333,335,337,345,348,350}.

H Independent response-centered Transformer audit

The method, candidate, and certificate seals have SHA-256 prefixes 2CB9738FD630392C, 43A016761DEB48CB, and 05D9FC835AEB58C8. The outcome audit prefix is E8BEEC6A28AC1AC3. The method seal precedes training seeds 355–378; the candidate seal precedes every randomized query; and the certificate seal precedes every future rollout.

Table 8: All 19 independently frozen response-centered candidates. “q” is the first issuing shared-probe power; “base” is the matched power-8 fixed-radius rule. Every finite centerline prediction equals the revealed lead.

Seed	Gate	Anchor	Lead	q	v3	Base	Disposition
360	70%	3,480	107	3	yes	yes	issued
361	70%	1,880	251	–	no	no	closure
361	80%	2,800	265	2	yes	yes	issued
365	70%	1,480	127	–	no	no	construction
366	70%	1,040	28	2	yes	yes	issued
366	80%	1,120	2	1	yes	yes	issued
366	90%	1,360	70	2	yes	yes	issued
368	80%	2,880	256	–	no	no	closure
369	70%	4,160	157	2	yes	yes	issued
369	80%	4,480	118	2	yes	yes	issued
369	90%	5,760	232	2	yes	yes	issued
370	70%	2,280	275	2	yes	yes	issued
372	70%	3,440	246	4	yes	no	issued
373	70%	1,280	247	–	no	no	closure
373	80%	1,760	214	–	no	no	closure
375	70%	920	217	–	no	no	early
375	80%	1,800	118	3	yes	yes	issued
378	70%	3,640	261	–	no	no	closure
378	90%	4,800	138	–	no	no	early

The 72-case disposition is 19 candidates, 42 trigger-visible screens with no future modal event, and 11 never-eligible gates. V3 issues 11/19 candidates (11/72 cases) across seven seeds; the fixed-radius baseline issues 10/19. Conditional coverage is 11/11 and 10/10, respectively. First issuing powers have counts (1, 7, 2, 1) for $q = (1, 2, 3, 4)$.

Seed 365 exposes an implementation invariant rather than a favorable exclusion. The blind selector found an event on a truncated modal path, while the constructor required every sweep to reach the full 300-step horizon. Before outcomes, a hash-bound execution amendment retained the case in the denominator as a zero-query abstention and prohibited retrying, shortening, or changing the numerical cap. Future selectors enforce the constructor’s full horizon invariant.

All 11 issued total-state radii contain the revealed trajectories. The stricter response-centered remainder diagnostic has 11 float64 violations at machine scale; the maximum absolute radius deficit is 3.07×10^{-14} . Every bracket survives added state-radius padding through 10^{-8} and all survive 10^{-7} (maximum width one), while only two survive 10^{-6} . This records numerical headroom but does not infer an outward error bound from observed discrepancies.

A separate 256-bit Arb audit treats each stored binary64 probe norm Y as an exact dyadic input, outward-encloses the folded-normal quantile and every $(Y/c_\delta)^{1/(2q)}$ root, and takes the maximum with each stored bound. The stored c_δ is already one ulp below the exact-quantile enclosure; 13,052 scalar rows require upward repair, with maximum factor $1 + 8.9 \times 10^{-16}$. All 11 brackets remain identical and minimum logic slack is 1.80×10^{-4} (audit SHA-256 prefix 74B40EE5E10C19CE). HVP/VJP, norm accumulation, jets, and margins are outside this conditional scalar audit.

The post-seal time-resolved theorem audit on seed 372 reduces the forcing term to 67.5% of the scalar value and the power-4 state radius from 2.73×10^{-14} to 2.06×10^{-14} . Combining that theorem with a role-stratified allocation of the unchanged 10^{-6} family budget closes the same [246, 246] event at power three. The audit file has SHA-256 prefix 6B1084D07558AC0A; it is method-development evidence and leaves the sealed v3 disposition unchanged.

I Cancellation-safe second-response audit

Corollary 7 was implemented after the v3 seal without reading future trajectories. The implementation contracts the third derivative of the objective twice with the parameter component of $\tilde{z}_j$, propagates the resulting signed $q^{(2)}$ through one more causal Green sweep, and reserves the local fourth-order Taylor bound plus arithmetic residuals in Eq. (54). It never subtracts two nearly equal optimizer maps.

The adaptive policy first applies the prospective zero-extra closure. If that issues, it stops; if $bp \geq 1$, no second-order forcing can repair the current power, so it refines the shared Gram iterate. Only when $bp < 1$ and the zero-extra forcing still fails does it construct $q^{(2)}$ and $\tilde{y}$, which are then reused at every later power.

Table 9: Outcome-blind post-seal second-response audit. Net HVPs subtract the extra causal responses but exclude directional third-product cost. The amplified benchmark includes every branch-specific fourth-order envelope. No prospective disposition or coverage count changes.

Quantity	Result
Green-evaluable sealed records	15
Zero-extra / directional closures	11 / 15
Prior abstentions converted	4
Existing closures advanced	5
Adaptive invocations / no-added-work cases	9 / 6
Progressive power levels removed	22
Net objective HVPs removed	167,250
Directional third products added	1,797
Local fourth-order Taylor gates	9/9
Minimum Taylor headroom	55.3×
Paired branch / next-power median	78.58 / 233.45 s
Paired speedup (four-pair range)	$2.91 \times (2.77\text{--}3.28 \times)$
Amplified secant λ / bracket	4096 / [28, 28]
Secant / third-AD / next-power median	6.56 / 8.10 / 26.01 s
Secant speedup vs. third-AD / next power	$1.19 \times / 4.04 \times$
Secant analytic headroom	24.4×
Outward scalar jets / precision	204 / 192 bits
Exact forcing bound / wall time	$1.31 \times 10^{-29} / 9.80$ min
Response-free bracket / forcing headroom	[28, 28] / $24.41 \times$
Relinearized mixed coefficient	$0.627 \rightarrow 0$
Relinearized 16-probe forcing headroom	$173.73 \times$
Four-probe bracket / forcing headroom	[28, 28] / $2.00 \times$
Relinearized Green applications	$16 \rightarrow 4$
Four- / 16-probe median time	6.51 / 23.64 s ($3.63 \times$)

At each invoked stopping power, the measured directional forcing is at most 2.60×10^{-13} of the zero-extra bound (median 3.02×10^{-15}). A separate deterministic scalar jet propagates Transformer derivatives through order four. Rather than charging a large anchor ball to every parameter block, it evaluates the exact segment ball $c_j + t\tilde{z}_j$ checkpointwise. All nine invoked cases satisfy the analytic Taylor gate; headroom over σ_2 ranges from $55.3 \times$ to 1.80×10^5 . The independent structural audit reproduces all counts, operator accounting, sealed Green domination, and fourth-order inequalities (SHA-256 prefix 75A8FAA37DFC92EF). The remaining headroom is not silently promoted to a proof: outward directional-product and recurrence residual bounds are still required before this post-seal audit can become a fresh computer-assisted Transformer certificate.

The incremental timing audit holds the centerline, first response, and first Gram power fixed, then alternates one additional 16-probe Gram power with the directional forcing plus scalar causal response. On sealed seed 366 at the 70% gate ($H = 52$), all four checksums agree and every pair favors the directional branch; the independent arithmetic/hash audit reads no future outcome. Absolute times were collected under concurrent workspace load, so we interpret the within-pair ratio rather than claim a machine-independent latency.

Corollary 9 was then implemented with the fixed power-of-two amplification $\lambda = 4096$. It evaluates one shifted gradient and one existing HVP per checkpoint, divides the nonlinear numerator by λ^2 , and reserves the exact ray-secant discrepancy from Eq. (60). On the same sealed $H = 52$ case it closes the power-one [28, 28] bracket, while the analytic discrepancy consumes only $1/24.4$ of the admissible injection-error budget; 6.01×10^{-20} remains for outward arithmetic and recurrence residuals. Three rotating-order triples time the secant, third-product, and next-power branches after identical shared setup. Their medians are 6.56, 8.10, and 26.01 seconds, respectively; every pair favors the secant over the next power (range $3.82\text{--}4.37 \times$). An independent arithmetic/hash audit reproduces the bracket, closure, headroom, checksums, and every timing ratio without reading a future outcome. After a

16-probe development audit, a new protocol froze four fresh forcing probes under a separate nonce. A 192-bit Arb multi-jet evaluates attention, exact GELU, softmax, cross-entropy, and weight decay while sharing each forward pass across all four directions. An adversarial audit caught a preliminary binary64 subtraction in $w = g_w - g_\theta$; v2 preserved every probe and formed that difference inside Arb. Across all 51 checkpoints, its 204 outward intervals bound the exact secant forcing by 1.31×10^{-29} , retain $[28, 28]$ with $24.41\times$ headroom, and take 9.80 minutes wall time (8.58 minutes for jets) on four workers. Twelve randomized architecture-level tests match PyTorch objectives, gradients, and mixed products within 6.67×10^{-16} ; an independent interval/hash audit re-sums every row. This closes the scalar secant obligation conditional on stored dyadic inputs, not the upstream Transformer arithmetic or prospective evidence boundary.

Corrected-path Green audit. We next froze a rebuilt Green operator on the corrected center $b = c + \tilde{z}$. The first protocol evaluated $G(b_j) - b_{j+1}$ literally in float64; cancellation inflated the measured forcing norm to 3.10×10^{-15} and produced a negative discriminant. We retain this failed interface. A second, separately frozen protocol instead used Eq. (72) with the amplified secant, so it never subtracts nearly equal optimizer maps. With 16 fresh probes at power one, it removes the old mixed coefficient 0.6273518, obtains $173.73\times$ injection-forcing headroom, and retains $[28, 28]$ without a second causal response.

That slack motivated one prespecified four-probe reduction, frozen before its new Gaussian block. The resulting Green bound is looser ($\bar{\kappa} = 45015.33$), yet the corrected closure retains $[28, 28]$ with $2.0028\times$ forcing headroom and a total radius 2.86×10^{-15} . An independent outcome-blind auditor regenerates all four Gaussian vectors, $\bar{K}_H^\top \bar{K}_H$ products, calibration, scalar root, margins, and bracket. The logical Green block falls from 16 to four applications. Three alternating matched pairs give medians 6.51 versus 23.64 seconds, or $3.63\times$ (pairwise range $3.48\text{--}4.10\times$), with shared centerline setup excluded. These post-seal audits do not alter the prospective cohort. The scalar secant forcing is outward conditional on stored dyadic inputs; response recurrence, rebuilt Green products, derivative envelopes, and output margins remain float64/high-confidence rather than a complete computer-assisted Transformer proof.

We then froze a cohort-wide release over all 15 Green-evaluable v3 records, before drawing its Gaussian block and without reading outcomes. It combines the corrected center, prefixes $m \in \{4, 8, 16\}$, one Gram power, and equal family spending. Every record issues the same bracket as the sealed directional baseline: 14 stop after four probes and one after eight. Relative to that baseline, the Green block contracts from 560 to 64 Gram applications ($8.75\times$); including the response/replay accounting declared by each protocol, the theoretical linearized schedule contracts from 1,150 to 144 sweeps ($7.99\times$). The minimum forcing headroom is $2.293\times$ and the minimum strict event-logic slack is 1.45×10^{-6} .

This is the complete pre-existing operator cohort, not a pass-selected subset. Each of the other four among 19 frozen candidates failed the original deterministic pre-query screen, so the frozen protocol instantiated no Green operator on which a probe-count replay could be defined.

The new Green family spends 10^{-6} and the inherited output family spends at most 10^{-6} , giving the explicit combined upper bound 2×10^{-6} . An independent auditor regenerates all 64 unique Gaussian vectors and recomputes every operator, root, closure, margin, hash, and bracket. Direct image screening under the same Gaussian event issues four cases before a transpose; Gram fallback issues the other 11. It therefore avoids 16 of 128 staged probe sweeps (128 to 112 , $1.143\times$), a useful but deliberately separate gain from the larger corrected-prefix reduction.

The scaled-momentum factorization in Corollary 12 permits a smaller operator without changing the reference path or output rule. A post-v1.0.1 protocol fixed $T_\theta = \Pi_\theta \bar{K}_H \mathcal{B}$, the same 4/8/16 direct/Gram policy, a fresh nonce, and a new 10^{-6} Green-family budget before drawing any probe. It preserves all 15 inherited brackets, with six direct-image and nine Gram releases, and reduces the staged total from 112 to 96 logical sweeps ($1.167\times$). An independent mechanical verifier checks all 17 sealed dependency hashes and 15 caches and recomputes every stored Gaussian bound, quadratic root, bracket, and aggregate.

An adversarial proof audit then made the causal indexing explicit: $\bar{K}_H$ returns $(e_1, \dots, e_H)$ while update j uses e_j . The earlier scalar bound remains valid because the shift in Eq. (76) has norm one. A separately sealed follow-up removed the identically zero update-zero forcing block as prescribed by Q_0 . All 15 brackets again survive, but the total remains 96 sweeps (median selected randomized

gain-bound ratio 0.994), so the prespecified strict-improvement gate fails. We report this negative result because it both validates the corrected indexing and locates the remaining bottleneck: coarse randomized release thresholds, not the zero anchor block.

Finally, Proposition 5 is executable rather than only an accounting identity. Three natural Transformer records reproduce the full- $H = 300$ centerline bitwise at sealed horizons 26, 131, and 299, including the same persistent-event prediction. Prefix streaming reduces centerline time by $9.94\times$, $2.09\times$, and $1.06\times$, and estimated path storage by $39.6\times$, $10.5\times$, and $4.84\times$, respectively.

Corollary 13 was then replayed on the same complete 15-case operator cohort. The deterministic screen alone issues eight cases with the same singleton brackets, while the existing output-probe path handles the other seven; staged issuance therefore remains 15/15. Those eight releases remove 1,432 randomized output-Jacobian operators and 22,912 $q = 1$ Gram applications. Their maximum analytic margin radius is 4.56×10^{-10} against minimum strict logic slack 1.80×10^{-4} , and their probability upper is only the 10^{-6} Green event. The audit reads zero outcome files. A separately implemented replay recomputes all analytic drift constants, quadratic roots, anchor margins, and event brackets and reproduces 8/15 exactly.

Table 10: Frozen, outcome-blind corrected-prefix release over every Green-evaluable v3 record. The staged direct/Gram audit reuses the same Gaussian event, so it adds no failure probability. These are post-seal mechanism and implementation results, not additional prospective coverage counts.

Quantity	Result
Evaluated / issued / preserved brackets	15 / 15 / 15
Stopping prefix $m = 4/8/16$	14 / 1 / 0
Green Gram applications	560 $\rightarrow$ 64 ($8.75\times$)
Declared linearized sweeps	1,150 $\rightarrow$ 144 ($7.99\times$)
Pairwise Green reduction, median (range)	$8\times$ ($4\text{--}16\times$)
Minimum forcing headroom	$2.293\times$
Direct-image / Gram-fallback routes	4 / 11
Staged probe sweeps	128 $\rightarrow$ 112 ($1.143\times$)
Full-state / parameter-forcing staged sweeps	112 $\rightarrow$ 96 ($1.167\times$)
Anchor-zero refinement versus parameter forcing	96 $\rightarrow$ 96 (gate failed)
Deterministic-jet / output-probe routes	8 / 7
Randomized output operators eliminated	1,432
Output $q = 1$ Gram applications eliminated	22,912
Streaming time reduction, $H = 26/131/299$	$9.94/2.09/1.06\times$
Streaming storage reduction, $H = 26/131/299$	$39.6/10.5/4.84\times$
Combined failure-probability upper bound	2×10^{-6}
Minimum strict event-logic slack	1.45×10^{-6}
Outcome files read	0

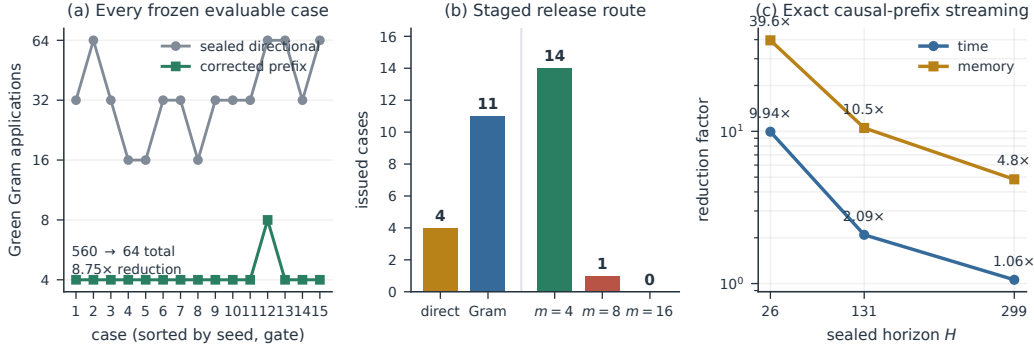


Figure 5: Practical effect of corrected-path, prefix-local release. (a) Every frozen evaluable case preserves its bracket with fewer Green Gram applications. (b) Direct images issue four cases before a transpose; most still require the sharper Gram release. (c) Causal streaming is bitwise identical to the full path and pays only for the sealed horizon.

The execution ledger preserves two deviations. An initial smoke execution double-counted a pending block, generated and applied two vectors, and then errored before producing any prefix statistic, result, or cache; that nonce was discarded and the corrected loop was frozen under a fresh nonce. In one v2 case, a later norm-only issuance short-circuited before a cached direct response was reused. This was conservative: the stored bound exceeds the latent direct-aware bound by 1.11×10^{-8} relatively, and no disposition, prefix, or bracket changes. Both events and their hashes remain in the supplement rather than being silently repaired.

J Directional block remainder and mixed-jet audit

The post-release audit froze the 15-case corrected-path cohort, disclosed seed 366 at the 80% gate as the development row, and reserved the other 14 cases for the promotion decision. It changed only the fourth-order Taylor term: centerlines, three-sweep corrections, domain radii, recorded family-calibrated Green bounds, drift constants, and event logic were inherited unchanged. No future-outcome file was read and no new Gaussian vector was drawn.

For the one-block Transformer, the analytic partition has 14 orthogonal blocks: joint token/position embedding; separate query, key, and value weight/bias blocks; attention output weight/bias; two feed-forward weight/bias pairs; and the readout. Nonnegative derivative polynomials are propagated through attention, GELU, softmax, and cross entropy. Segment values are inflated by a monotone first-order fixed point before the fourth-order majorant is evaluated. Product/chain-rule scalarization, exact partition, Euler homogeneity, and 12 direct mixed-fourth autodiff tests all pass.

Table 11: Frozen directional-block audit. “New” means closure under the three-sweep rule where the scalar fourth-order comparator abstains; all event brackets already belong to the sealed four-sweep cohort and are not counted as new prospective coverage.

Quantity	Result
Cases / nondevelopment cases	15 / 14
Directional/scalar ratio, holdout median	3.45×10^{-4}
Holdout tightening, median (range)	$2,899 \times (977-10,997 \times)$
Three-sweep closures, scalar / directional	1 / 4
New nondevelopment closures	3
Issued / sealed bracket retained	4/4 / 4/4
Issued offsets	2, 28, 70, 118
Minimum strict event slack	1.36×10^{-3}
Every checkpoint no weaker than scalar	yes
Independent mixed-jet maximum relative error	4.19×10^{-15}
Polynomial / mixed-jet median runtime	154.31 / 115.92 s
Whole-case runtime ratio / frozen gate	$1.33 \times / 2 \times$ (failed)
Outcome files / new randomized queries	0 / 0

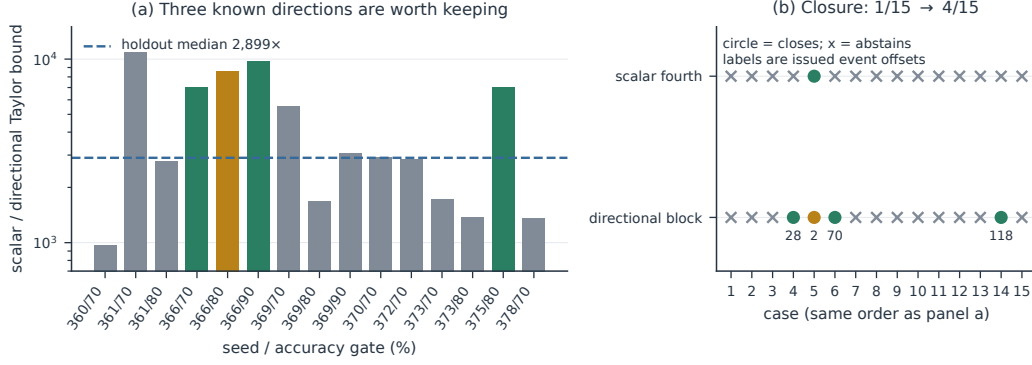


Figure 6: Directional fourth-order geometry in the complete frozen panel. (a) Scalar-to-directional Taylor tightening; green bars are new holdout closures, ochre is the disclosed development row, and gray cases still abstain. (b) The scalar rule closes one case, whereas the directional rule closes four and returns the labeled sealed offsets.

The independent implementation does not differentiate the stored polynomial. It propagates a bivariate jet in a known variable t and a free block variable ε , retaining D_t^k and D_ε^k for $k \leq 3$. Across all 15 cases it reproduces every local Taylor bound and closure decision. Its prespecified $2\times$ whole-case speed gate fails at $1.33\times$ because centerline replay remains dominant. This negative timing gate is part of the result: the practical saving established here is one fewer predeclared centerline sweep on four records, not a twofold acceleration of the complete verifier.

K Matched unsigned baseline and sweep audit

The unsigned comparator keeps the same causal Green operator and therefore the same finite-window conditioning information. It changes only the zero-order response:

$$Z = \|K_H s\|_2 \longrightarrow Z_u = \kappa \|s\|_2. \quad (119)$$

For its own ball it solves the radii polynomial

$$p_u(R) = Z_u + \frac{1}{2}\kappa M(R)R^2 - R \leq 0 \quad (120)$$

and chooses the smallest admissible root. The analytic neural envelope is coordinatewise nondecreasing in R . Thus, whenever $Z_u > 2Z$, every admissible unsigned radius contains the frozen signed ball, and evaluating the unsigned closure with the smaller signed-ball $M(2Z)$ is already favorable to the baseline. Failure of that lower envelope proves failure at every admissible unsigned radius. This argument eliminates 17/18 matched cases without numerical reconstruction. The remaining case, seed 345 at the 80% gate, is reconstructed with its own outer-ball jet and retains [30, 30]; it is the baseline's sole issued certificate.

Table 12: Complete 0–4-sweep audit on the three sealed burned-development candidates. Defect is the median across candidates of the maximum scaled path defect.

Sweeps	Cumulative HVPs	Exact clocks	Median error	Defect
0	300	0/3	3	8.57×10^{-3}
1	600	3/3	0	2.23×10^{-3}
2	900	3/3	0	5.85×10^{-4}
3	1,200	3/3	0	2.58×10^{-5}
4	1,500	3/3	0	5.94×10^{-8}

The median incremental construction times for stages zero through four are 26.56, 24.58, 31.04, 34.70, and 37.15 seconds. These are diagnostic timings, not a fresh model-selection study. Their role is to expose the numerical mechanism behind the prospectively fixed choice of four sweeps.

L Matrix-free scaling accounting

For $H = 300$, four sweeps require 1,500 objective HVPs. The frozen $m = 16, q = 8$ Green probe requires 76,800 objective HVPs, and the output-margin probe requires 38,528 output-Jacobian Gram products. These are logical calls: the batched implementation evaluates the same 16 independent rows together, preserving their generation order, Y maximum, calibration, and union budget. Tests compare HVP, optimizer JVP/VJP, causal Green, transpose, output-Gram, and the final probabilistic bound against scalar execution; maximum relative error is 3.24×10^{-16} .

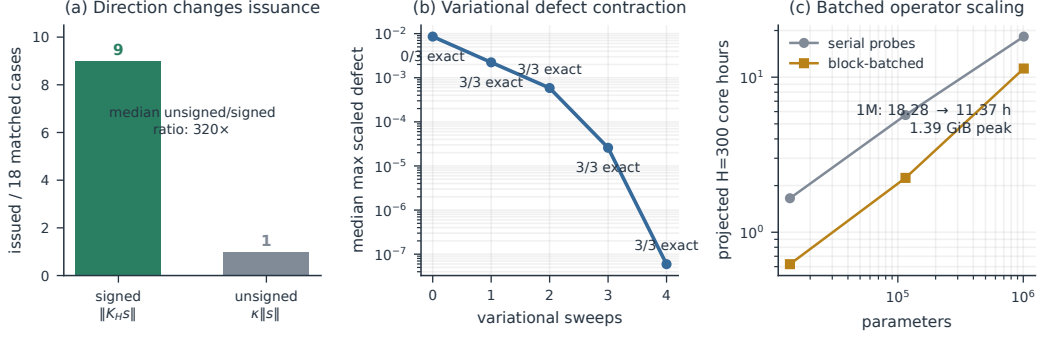


Figure 7: Mechanism and scale. (a) The matched scalar Euclidean unsigned finite-window baseline retains only 1/9 signed Transformer certificates; direction removal inflates the zero-order term by a median $320\times$. (b) Variational sweeps rapidly contract the known path defect; one sweep already makes all three burned-development clocks exact. (c) Evaluating the same 16 probes as one reverse-mode block lowers the matched $H = 300$ core projection at every tested width.

Table 13: Matched serial-versus-block-batched projection for the unchanged full $H = 300$, four-sweep, $m = 16, q = 8$ core. Both columns use same-run timings and exclude jets, orchestration, I/O, and cache effects.

Params	Serial core	Batched core	Speedup	Peak GiB
13,792	1.66 h	0.62 h	$2.66\times$	0.36
115,104	5.70 h	2.24 h	$2.54\times$	0.53
1,008,864	18.28 h	11.37 h	$1.61\times$	1.39

Table 13 uses serial and batched block timings from the same run. An earlier isolated-one-product multiplication yielded 6.16/18.98/65.88 hours; the JSON record retains it for traceability, but it is not used as the speedup denominator because same-run serial blocks are materially faster. Both projections exclude jets, Python orchestration, checkpoint I/O, and cache effects. At 1,008,864 parameters the batched 16-direction HVP and output-Gram blocks are $1.58\times$ and $1.79\times$ faster, with 1.39 GiB process peak.

For a direct wall-clock baseline, the historical frozen fixed- $q = 8$ paper batch has 22 candidate constructions taking 19.44 aggregate hours and a median of 58.02 minutes, whereas continuation of 300 actual updates estimated from all 24 training traces takes a median of 5.31 seconds ($655.8\times$ less). Those numbers describe the original implementation, not the composed verifier. On the sealed seed-366 80% case, three separately launched executions combine exact prefix streaming, one signed corrected-path sweep, deterministic neural jets, and a four-probe direct-image release. All reproduce [2, 2] in 9.01–11.71 seconds (median 9.21), with four forward Green applications, zero transpose applications, zero randomized output operators, and family failure upper 10^{-6} . Relative to the historical cross-batch median this is an $378\times$ reduction. A same-machine three-repeat control measures direct continuation at 0.298 seconds for the required 26 updates and 3.718 seconds for 300, leaving certification $30.9\times$ and $2.48\times$ slower. Continuation computes the future outcome; GREENCERT constructs an outcome-independent proof object before that trajectory is opened. On the small exact-real WDBC audit, where validated continuation is the matched proof baseline rather

than an ordinary rollout, the direction reverses: 7.12 hours for direct continuation versus 33.88 minutes for the GreenCert records (Table 4).

Two end-to-end replays rebuild all four sweeps, analytic envelopes, output logic, and the Green probe from the sealed checkpoint and committed streams. Seed 333’s [28, 28] certificate falls from 17.82 to 5.94 minutes ($3.00\times$); seed 345’s [30, 30] falls from 17.34 to 6.66 minutes ($2.60\times$). Maximum disagreement in any bound or closure statistic is 2.14×10^{-16} , and both dispositions are unchanged.

The witness-sparse audit reconstructs every one of the 11 response-centered brackets from its K certified success times and a greedy minimum-cardinality set of certified failures. Event-query times fall by a median $4.73\times$ (range $1.00\text{--}8.31\times$). Replacing the current all-289-input output operator by transition-state training operators (173 examples) and witness-state certification operators (58 examples) gives the exact example-level accounting $(H - 1)173 + |W|58$, a median 36.3% reduction relative to $H \cdot 289$. The predictable acquisition policy then discovers all 11 witness sets from raw centerline slacks and prior query results, reducing 1,882 full-time queries to 349 (audit SHA-256 prefix 1AEA107F9EFF5C5). A matched timing panel at horizons 26, 52, 94, and 142 preserves all four sealed brackets without fallbacks. Direct-sum role fusion reduces aggregate example-pair work 33.7% and gives $1.37\times$ operator and $1.35\times$ output-wall speedups (range $1.19\text{--}1.40\times$); including shared centerline time gives $1.11\times$. The independent panel audit has SHA-256 prefix E35B1DD723371F86.

A first monolithic replay composes online Green stopping with role-fused output transport on the sealed $H = 26$ case. It keys witnesses to the frozen centerline event, stops both operator families at $q = 1$, uses no fallback, and retains [2, 2]. Pair work falls 22.37% and the same-process output-operator wall time improves $1.29\times$; the matched complete replay improves only $1.018\times$ (95.79 seconds measured), because centerline construction and the Green solve dominate this short horizon. This directly identifies the remaining systems bottleneck rather than extrapolating output-only speedups (record SHA-256 prefix 006CC711EDA43CC4).

The final composition removes those two bottlenecks rather than combining their isolated speedup factors. Prefix streaming constructs only the required 26 checkpoints; corrected-path direct-image release stops after four forward queries; and Corollary 13 replaces every randomized output query by the already required ball-valid jets. Median phase times over the three fresh processes are 5.226 s (centerline), 1.119 s (signed correction), 0.678 s (jets), 2.007 s (Green), and 0.082 s (closure and event logic), with 9.206 s median end to end. The independent audit record has SHA-256 prefix 664993E294AD6010.

Proposition 3 also yields an explicit numerical contract. On that same $q = 1$ replay, simultaneous verified Gram residuals $\xi \leq 0.5999\tilde{Y}$ for every Green/output query still retain the singleton bracket (operator-bound inflation $1.2649\times$). Green-only and output-only tolerances are 4.2295 and 0.9107 times their terminal norms. These are admissible-residual sensitivities, not estimates of floating-point error. A 50–50 max-column/chi-block confidence split additionally reduces the state radius from 2.1331×10^{-15} to 2.0705×10^{-15} from the same probe block and with no additional Gram application.

An actual mixed-precision replay then evaluates every committed $q = 1$ Green/output Gram product in float32 and measures its discrepancy against the binary64 target. The largest observed residual ratio is 3.36×10^{-7} , 1.79×10^6 below the common admissible threshold; the residual-corrected outward scalar roots retain [2, 2] with only $1 + 2.75 \times 10^{-8}$ Green-bound inflation. Across four separately launched five-repeat alternating-order runs, all 20 paired $q=1$ kernel speedups exceed one; their pooled median is $1.83\times$ (range $1.04\text{--}2.38\times$; invocation medians 1.66/1.86/1.89/1.78 $\times$). These discrepancies are measured rather than outward-enclosed, so the replay demonstrates concrete mixed-precision headroom, not a completed exact-real neural-kernel proof (aggregate-audit SHA-256 prefix 7E5619D0B9EE6A41).

Finally, the output-recentering corollary is audited without opening any future trajectory. Across the 15 Green-evaluable response-centered records, its proof-preserving intersection with the original enclosure retains 11/11 issued certificates at identical powers and brackets. The recentered output radii have maximum ratio 0.091–0.602 at issuance (median 0.365), with no added random query. No abstention converts and no power decreases on this cohort; the measured gain is strictly smaller output uncertainty and precision headroom.

For architectural transport, a separate profile uses two pre-LayerNorm blocks, 102,400 parameters, and a standard bias-corrected AdamW state of dimension 307,200. The analytically derived optimizer JVP/VJP uses one neural HVP and agrees with full autograd to below 9×10^{-18} on an explicit test; optimizer and horizon-3 Green adjoint errors at profile scale are 5.45×10^{-15} and 4.40×10^{-15} . This establishes matrix-free linear transport only, not the global LayerNorm/AdamW jets needed for closure.

M Dense outward and disjoint HVP details

In the dense mod-11 study, all eight fresh models fit training. Among 40 seed-gate pairs, 37 events occur and the chronological rule selects 27 anchors. The uncorrected v1 tube issues 6/27 certificates; one-sweep recentered v2 issues 16/27, loses none of the six, and contains all 16 crossings. Median capped state horizon moves from 26 to 174 updates; 13/27 recentered tubes reach the full 250-step window. Fourteen brackets are singletons; the other two are $[18, 20]$ and $[64, 70]$, containing leads 18 and 66. All issued brackets survive a separate 192-bit outward pass unchanged.

The mod-13 HVP study uses widths 48, seeds 202–205, and disjoint train/trigger/certification sets of 101/34/34. Of 20 seed-threshold cases, 19 become eligible; 13 gates are already present, four screens find no future candidate, and two candidates remain. Both issue persistent singleton brackets, $[31, 31]$ and $[4, 4]$, and both cover. They come from one seed. No dense Hessian entry is formed; the two-radius uncertainty certificate uses a rank-64 active subspace and a declared family-wise probe failure probability 4.2×10^{-7} .

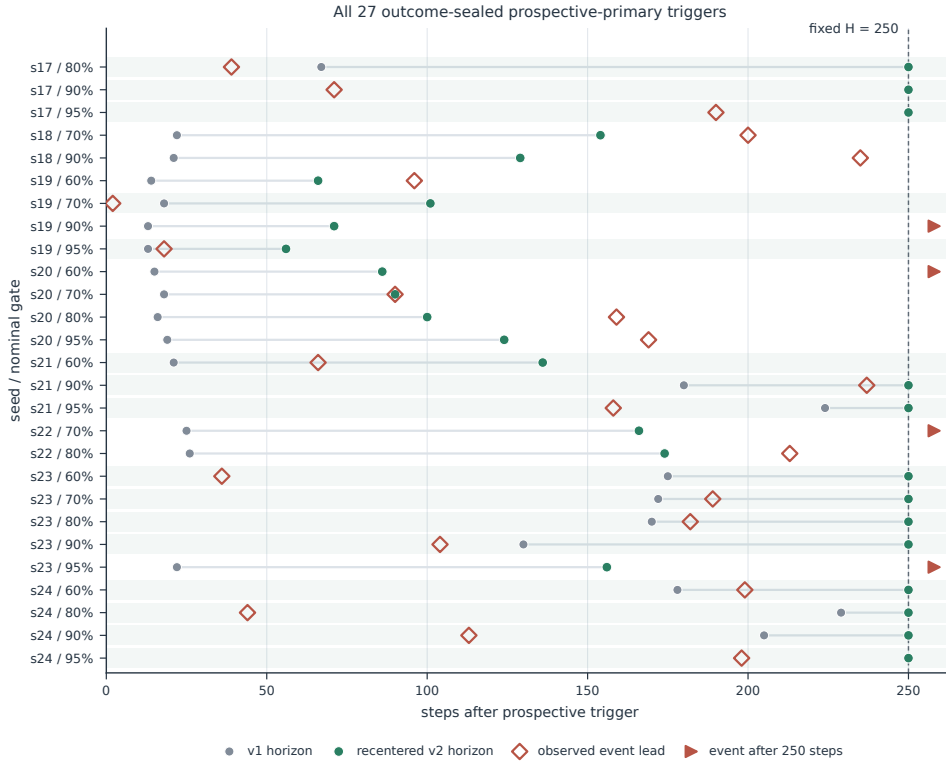


Figure 8: All 27 outcome-sealed dense triggers. Signed recentering (teal) extends the median capped rigorous horizon from 26 to 174 updates relative to the norm-only baseline (gray). Diamonds are observed event leads; triangles lie beyond the fixed 250-step window.

N Numerical and reproducibility record

Fixed-radius Transformer chain. The Transformer method, candidate manifest, pre-outcome certificate seal, and final aggregate have SHA-256 prefixes 14EABFDE3823E228, D61FDFA72DAC8D70, 4B6EBBAA877280F6A, and A3D480FB92BFEC89, respectively. Complete hashes and a manifest of every claim-bearing artifact are included in the supplement.

An independent checker does not import aggregation helpers. It reloads all 23 certificate and post-seal audit files, verifies their hashes, recomputes $R = 2Z$, both closure inequalities, every 25-step persistent bracket from the guaranteed/possible count arrays, the randomized budget, all coverage fields, and seed-clustered summaries. All checks pass. Unit tests compare causal Green products with an explicit block matrix, verify the real Transformer adjoint, check the Gram exponent $1/(2q)$, finite-difference the scaled momentum JVP, and enforce that candidate screening cannot read outcomes or make probabilistic queries. Random-direction jet tests are debugging checks only; Appendix B provides the line-by-line global analytic derivation. A separate exact-graph regression verifies that the sealed bias-free-readout relaxation adds exactly one nonnegative first-degree monomial and no second- or third-degree term.

Response-centered Transformer chain. The v3 method, candidate, certificate, and outcome-audit SHA-256 prefixes are 2CB9738FD630392C, 43A016761DEB48CB, 05D9FC835AEB58C8, and E8BEEC6A28AC1AC3. The single construction exception is separately bound by pre-outcome amendment seal 66C452C64FBBBA75. Independent aggregation verifies all 72 dispositions, 19 modal predictions, 11 issued brackets, the matched 10-case baseline, all first-issuing powers, 3,649 queried operators, and the realized union bound. Exact nonlinear scalar tests cover the response-centered and time-resolved quadratic roots; resumable Gram tests reproduce every stored power row from the same committed probe state.

WDBC chain. The complete hashes appear in Appendix E. Its selector test attempts forbidden certification-set accesses and requires each to fail. Candidate and certificate seals are verified before the independent outcome join. A second auditor that does not import the runner’s aggregation code reconstructs all 72 dispositions and all 56 brackets. The direct-baseline audit additionally checks that the outward recurrence contains no Green, κ , probe, or random variable and verifies all 40 cached zero-radius anchor tubes.

Digits chain. The raw scikit-learn arrays, protocol code, fresh seed list, and all constants are committed by method seal 9901CD1A0E3FEABF. Loader tests prove that training/selection cannot return certification tensors. Candidate seal BFA31D77050F164D fixes all 24 coordinates; certificate seal 99AC9C628AFB4FC9 records 7/6/1 signed/unsigned/signed-only issuance before outcome reconstruction. A separate auditor that does not import the runner recomputes all 24 anchors, checks every seal and event, and confirms 7/7 coverage; checkpoint reconstruction error is zero. A second independent audit verifies all seven 192-bit direct tubes, cache hashes, positive logic slacks, exact bracket identity, and the signed-only [147, 147] event. A post-seal selectivity auditor independently identifies the two inaccurate finite centerlines and verifies that both records are deterministic early abstentions with no randomized query.

Post-seal hardening chain. The batched-operator tests compare every matrix-free product with scalar evaluation. Two complete replay records point to immutable original certificate hashes and reproduce their centerlines, bounds, closures, and brackets. A separate aggregator recomputes all timing ratios and million-parameter projections, verifies the digits outward caches without importing their runner, and checks source hashes and adjoint identities for the LayerNorm/AdamW primitive audit. The matched anytime benchmark binds the original certificate and script hashes, reproduces the [2, 2] decision with zero trace deviation, and records both online and forced-q8 executions. The time-resolved/role-budget audit uses only sealed v3 traces and explicitly marks the prospective issuance result unchanged. The witness-sparse audit reconstructs all 11 issued first-passage brackets, and 37,000 brute-force interval instances verify the greedy minimum witness algorithm. A second policy test covers 15,875 exhaustive and 2,000 randomized acquisition cases; its v3 replay reconstructs all 11 brackets using only predictable queries. Candidate-generic independent role-fusion audits bind each sealed certificate, centerline, benchmark, and complete cache; they reproduce four brackets with zero trace discrepancy and no fallback. The panel aggregator recomputes query counts, pair

work, partial-phase timings, and the naive-versus-fused comparison without importing the benchmark runner. The 256-bit Arb audit encloses calibration and scalar roots conditional on stored Y ; an independent script rechecks its exact interval, certificate hashes, and 11 retained brackets. A strict post-fixed Transformer wrapper has a fast anchor-state regression and a complete 26-state audit on the shortest issued case; it removes every literal binary64 post-fixed deficit while reproducing the frozen bracket (audit SHA-256 prefix `4FBFC4926ABD65BE`). The inexact-operator tests separately check 6,000 randomized Gram cases, 2,405 operator-cap precision cases, 1,280 causal response-residual cases, and 960 inexact variational sweeps. Four independently audited float32/binary64 replays bind the immutable combined benchmark, rerun all outward scalar supersolution checks, and recover the same bracket in every invocation; all 20 paired kernel speedups exceed one, with pooled median $1.83\times$.

Exact-real boundary. The 192-bit WDBC, digits, and dense-modular passes treat stored binary64 checkpoints as exact inputs and outwardly enclose network, Hessian, eigensolver, recurrence, and margin arithmetic. They certify continuation from those checkpoints, not the PyTorch/BLAS training process that produced them. The WDBC/digits direct recurrences independently validate issued brackets rather than every randomized Green inequality. The Transformer neural-operator layer remains float64/ideal-PRNG; its outward scalar calibration/root audit is conditional on stored Y . Accordingly, those Green claims are exact-Gaussian propositions with deterministic-PRNG implementations, not end-to-end computer-assisted proofs (Appendix B).

Modular data are exhaustive; WDBC and bundled digits carry source checksums and persistent UCI identifiers. The supplement provides source, tests, frozen protocols and seals, manifests, compact outcomes, independent audits, and exact reproduction commands; large checkpoint arrays are regenerated.